

Cape Romano-Ten Thousand Islands Aquatic Preserve

SEACAR Habitat Analyses

Last compiled on 10 June, 2026

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Funding & Acknowledgements

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This report was funded in part, through a grant agreement from the Florida Department of Environmental Protection, Florida Coastal Management Program, by a grant provided by the Office for Coastal Management under the Coastal Zone Management Act of 1972, as amended, National Oceanic and Atmospheric Administration. The views, statements, findings, conclusions and recommendations expressed herein are those of the author(s) and do not necessarily reflect the views of the State of Florida, NOAA or any of their sub agencies.

Published: 2026-06-10



Threshold Filtering

Threshold filters, following the guidance of Florida Department of Environmental Protection's (*FDEP*) Division of Environmental Assessment and Restoration (*DEAR*) are used to exclude specific results values from the SEACAR Analysis. Based on the threshold filters, Quality Assurance / Quality Control (*QAQC*) Flags are inserted into the *SEACAR_QAQCFlagCode* and *SEACAR_QAQC_Description* columns of the export data. The *Include* column indicates whether the *QAQC* Flag will also indicate that data are excluded from analysis. No data are excluded from the data export, but the analysis scripts can use the *Include* column to exclude data (1 to include, 0 to exclude).

Table 1: Continuous Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Dissolved Oxygen	mg/L	-0.000001	50
Dissolved Oxygen Saturation	%	-0.000001	500
Fluorescent Dissolved Organic Matter	QSE	-	-
Salinity	ppt	-0.000001	70
Specific Conductivity	mS/cm	-0.000001	200
Turbidity	NTU	-0.000001	4000
Water Temperature	Degrees C	-5.000000	45
pH	None	2.000000	14

Table 2: Discrete Water Quality threshold values

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Ammonia, Un-ionized (NH3)	mg/L	-	-
Ammonium (NH4)	mg/L	-	-

<i>Parameter Name</i>	<i>Units</i>	<i>Low Threshold</i>	<i>High Threshold</i>
Chlorophyll a, Corrected for Pheophytin	ug/L	-	-
Chlorophyll a, Uncorrected for Pheophytin	ug/L	-	-
Colored Dissolved Organic Matter	PCU	-	-
Dissolved Oxygen	mg/L	-0.000001	25
Dissolved Oxygen Saturation	%	-0.000001	310
Fluorescent Dissolved Organic Matter	QSE	-	-
Light Extinction Coefficient	m ⁻¹	-	-
NO2+3, Filtered	mg/L	-	-
Nitrate (NO3)	mg/L	-	-
Nitrite (NO2)	mg/L	-	-
Nitrogen, inorganic	mg/L	-	-
Nitrogen, organic	mg/L	-	-
Phosphate, Filtered (PO4)	mg/L	-	-
Salinity	ppt	-0.000001	70
Secchi Depth	m	0.000001	50
Specific Conductivity	mS/cm	0.005000	100
Total Ammonia (N)	mg/L	-	-
Total Kjeldahl Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Nitrogen	mg/L	-	-
Total Phosphorus	mg/L	-	-
Total Suspended Solids	mg/L	-	-
Turbidity	NTU	-	-
Water Temperature	Degrees C	3.000000	40
pH	None	2.000000	13

Table 3: Quality Assurance Flags inserted based on threshold checks listed in Table 1 and 2

<i>SEACAR QAQC Description</i>	<i>Include</i>	<i>SEACAR QAQCFlagCode</i>
Exceeds maximum threshold	0	2Q
Below minimum threshold	0	4Q
Within threshold tolerance	1	6Q
No defined thresholds for this parameter	1	7Q

Value Qualifiers

Value qualifier codes included within the data are used to exclude certain results from the analysis. The data are retained in the data export files, but the analysis uses the *Include* column to filter the results.

STORET and WIN value qualifier codes

Value qualifier codes from *STORET* and *WIN* data are examined with the database and used to populate the *Include* column in data exports.

Table 4: Value Qualifier codes excluded from analysis

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>MDL</i>	<i>Description</i>
STORET-WIN	H	0	0	Value based on field kit determination; results may not be accurate
STORET-WIN	J	0	0	Estimated value
STORET-WIN	V	0	0	Analyte was detected at or above method detection limit
STORET-WIN	Y	0	0	Lab analysis from an improperly preserved sample; data may be inaccurate

Discrete Water Quality Value Qualifiers

The following value qualifiers are highlighted in the Discrete Water Quality section of this report. An exception is made for **Program 476** - *Charlotte Harbor Estuaries Volunteer Water Quality Monitoring Network* and data flagged with Value Qualifier **H** are included for this program only.

H - Value based on field kit determination; results may not be accurate. This code shall be used if a field screening test (e.g., field gas chromatograph data, immunoassay, or vendor-supplied field kit) was used to generate the value and the field kit or method has not been recognized by the Department as equivalent to laboratory methods.

I - The reported value is greater than or equal to the laboratory method detection limit but less than the laboratory practical quantitation limit.

Q - Sample held beyond the accepted holding time. This code shall be used if the value is derived from a sample that was prepared or analyzed after the approved holding time restrictions for sample preparation or analysis.

S - Secchi disk visible to bottom of waterbody. The value reported is the depth of the waterbody at the location of the Secchi disk measurement.

U - Indicates that the compound was analyzed for but not detected. This symbol shall be used to indicate that the specified component was not detected. The value associated with the qualifier shall be the laboratory method detection limit. Unless requested by the client, less than the method detection limit values shall not be reported

Systemwide Monitoring Program (SWMP) value qualifier codes

Value qualifier codes from the *SWMP* continuous program are examined with the database and used to populate the *Include* column in data exports. *SWMP* Qualifier Codes are indicated by *QualifierSource=SWMP*.

Table 5: SWMP Value Qualifier codes

<i>Qualifier Source</i>	<i>Value Qualifier</i>	<i>Include</i>	<i>Description</i>
SWMP	-1	1	Optional parameter not collected
SWMP	-2	0	Missing data
SWMP	-3	0	Data rejected due to QA/QC
SWMP	-4	0	Outside low sensor range
SWMP	-5	0	Outside high sensor range
SWMP	0	1	Passed initial QA/QC checks
SWMP	1	0	Suspect data
SWMP	2	1	Reserved for future use
SWMP	3	1	Calculated data: non-vented depth/level sensor correction for changes in barometric pressure
SWMP	4	1	Historical: Pre-auto QA/QC
SWMP	5	1	Corrected data

Water Column

The water column habitat extends from the water's surface to the bottom sediments, and it's where fish, dolphins, crabs and people swim! So much life makes its home in the water column that the health of marine and coastal ecosystems, as well as human economies, depend on the condition of this vulnerable habitat. Local patterns of rainfall, temperature, winds and currents can rapidly change the condition of the water column, while global influences such as [El Niño/La Niña](#), large-scale fluctuation in sea temperatures and climate change can have long-term effects. Inputs from the prosperity of our day-to-day lives including farming, mining and forestry, and emissions from power generation, automobiles and water treatment can also alter the health of the water column. Acting alone or together, each input can have complex and lasting effects on habitats and ecosystems.

SEACAR evaluates water column health with several essential parameters. These include nutrient surveys of nitrogen and phosphorus, and water quality assessments of salinity, dissolved oxygen, pH, and water temperature. Water clarity is evaluated with Secchi depth, turbidity, levels of chlorophyll a, total suspended solids, and colored dissolved organic matter. Additionally, the richness of nekton is indicated by the abundance of free-swimming fishes and macroinvertebrates like crabs and shrimps.

Seasonal Kendall-Tau Analysis

Indicators must have a minimum of five to ten years, depending on the habitat, of data within the geographic range of the analysis to be included in the analysis. Ten years of data are required for discrete parameters, and five years of data are required for continuous parameters. If there are insufficient years of data, the number of years of data available will be noted and labeled as "insufficient data to conduct analysis". Further, for the preferred Seasonal Kendall-Tau test, there must be data from at least two months in common across at least two consecutive years within the RCP managed area being analyzed. Values that pass both of these tests will be included in the analysis and be labeled as *Use_In_Analysis* = **TRUE**. Any that fail either test will be excluded from the analyses and labeled as *Use_In_Analysis* = **FALSE**. The points for all Water Column plots displayed in this section are monthly averages. Trend significance will be denoted as "Significant Trend" (when $p < 0.05$), or "Non-significant Trend" (when $p \geq 0.05$). Any parameters with insufficient data to perform Seasonal Kendall-Tau test will have their monthly averages plotted without a corresponding trend line.

Water Quality - Discrete

The following files were used in the discrete analysis:

- *Combined_WQ_WC_NUT_Chlorophyll_a_corrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Chlorophyll_a_uncorrected_for_pheophytin-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Colored_dissolved_organic_matter_CDOM-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Dissolved_Oxygen_Saturation-2026-May-18.txt*
- *Combined_WQ_WC_NUT_pH-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Salinity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Secchi_Depth-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Nitrogen-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Phosphorus-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Total_Suspended_Solids_TSS-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Turbidity-2026-May-18.txt*
- *Combined_WQ_WC_NUT_Water_Temperature-2026-May-18.txt*

Chlorophyll a, Corrected for Pheophytin - Discrete

Seasonal Kendall-Tau Trend Analysis

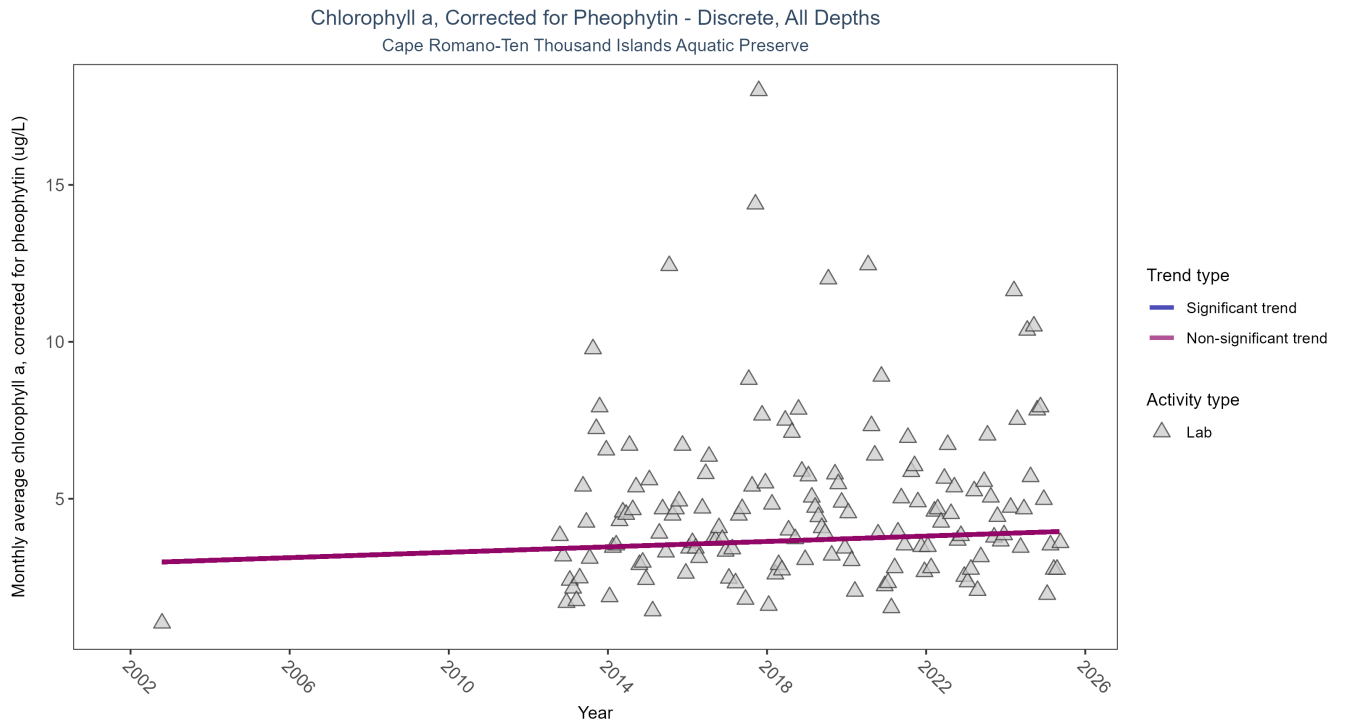


Figure 1: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 6: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Corrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	639	15	2002 - 2025	3.7	0.0914	2.9508	0.043	0.1463

Chlorophyll a, corrected for pheophytin, showed no detectable trend between 2002 and 2025.

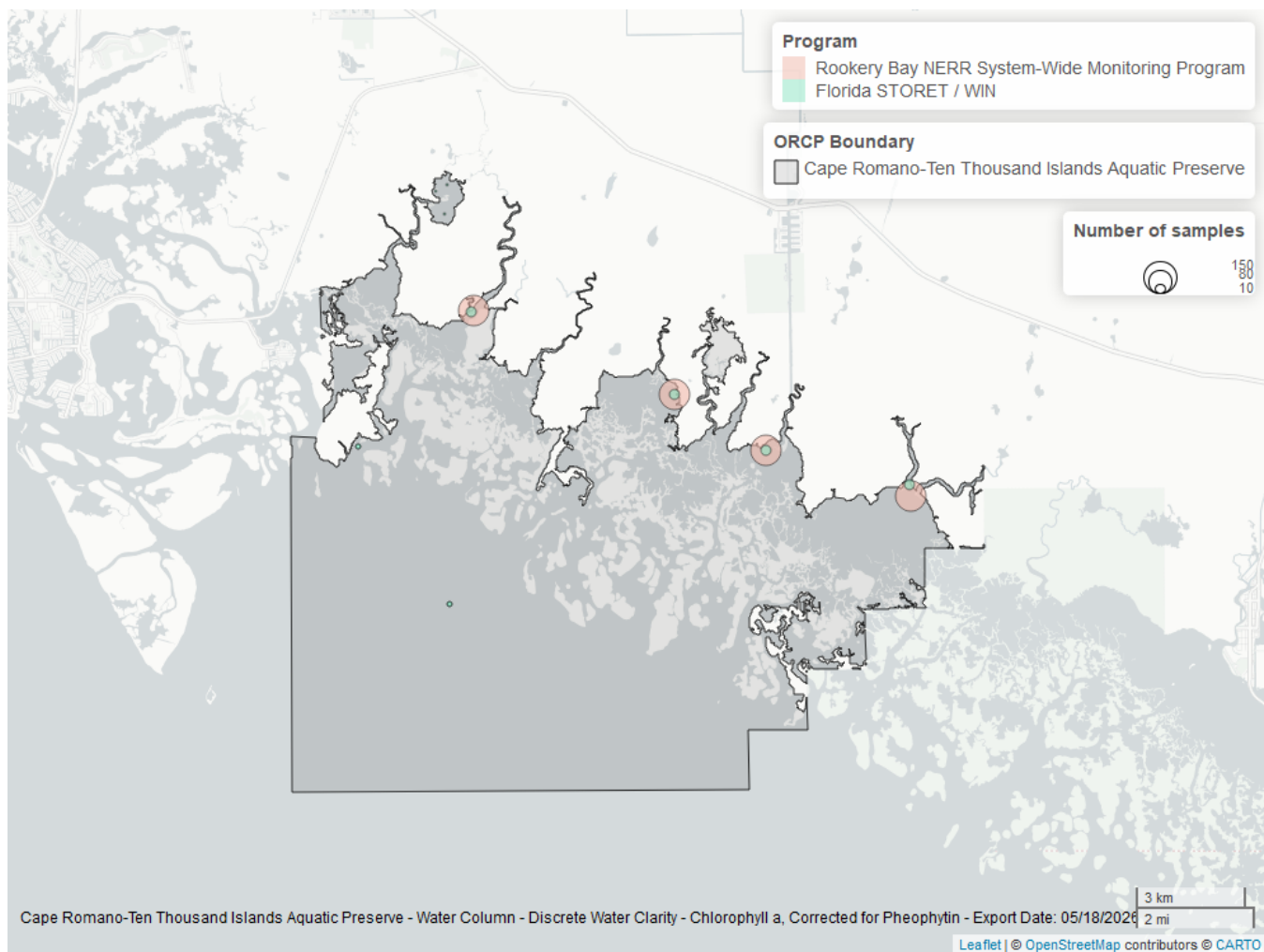


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 7: Programs contributing data for Chlorophyll a, Corrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
354	583	2012	2025
5002	68	2002	2021

Program names:

[354](#) - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹
5002 - Florida STORET / WIN²

Chlorophyll a, Uncorrected for Pheophytin - Discrete Seasonal Kendall-Tau Trend Analysis

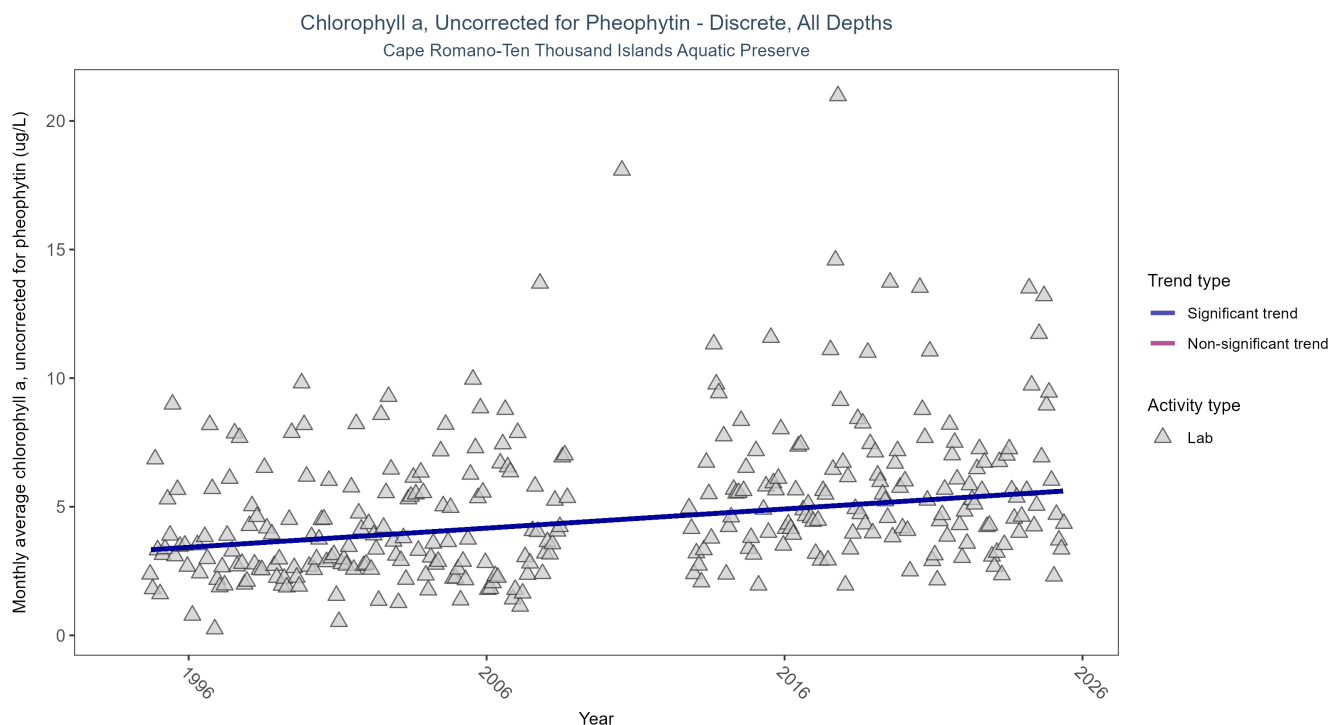


Figure 3: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 8: Seasonal Kendall-Tau Trend Analysis for Chlorophyll a, Uncorrected for Pheophytin

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2173	30	1994 - 2025	3.6567	0.2738	3.2739	0.0745	0

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.07 $\mu\text{g/L}$ per year, indicating a decrease in water clarity.

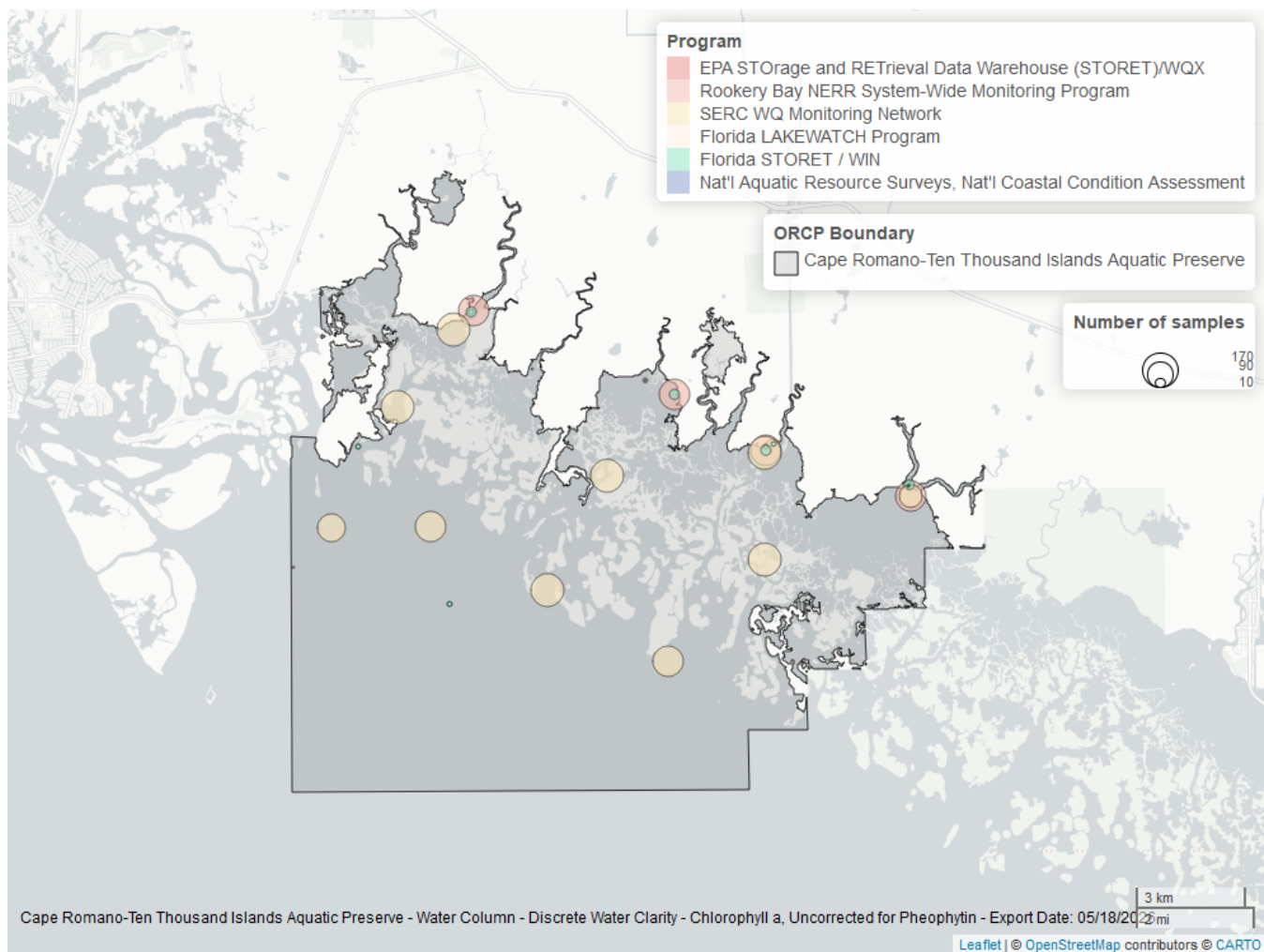


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 9: Programs contributing data for Chlorophyll a, Uncorrected for Pheophytin

ProgramID	N_Data	YearMin	YearMax
509	1512	1994	2008
354	583	2012	2025
5002	72	2001	2021
514	15	2001	2001
118	1	2010	2010
103	1	2015	2015

Program names:

509 - SERC Water Quality Monitoring Network³

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

5002 - Florida STORET / WIN²

514 - Florida LAKEWATCH Program⁴

118 - National Aquatic Resource Surveys, National Coastal Condition Assessment⁵

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Dissolved Oxygen - Discrete

Seasonal Kendall-Tau Trend Analysis

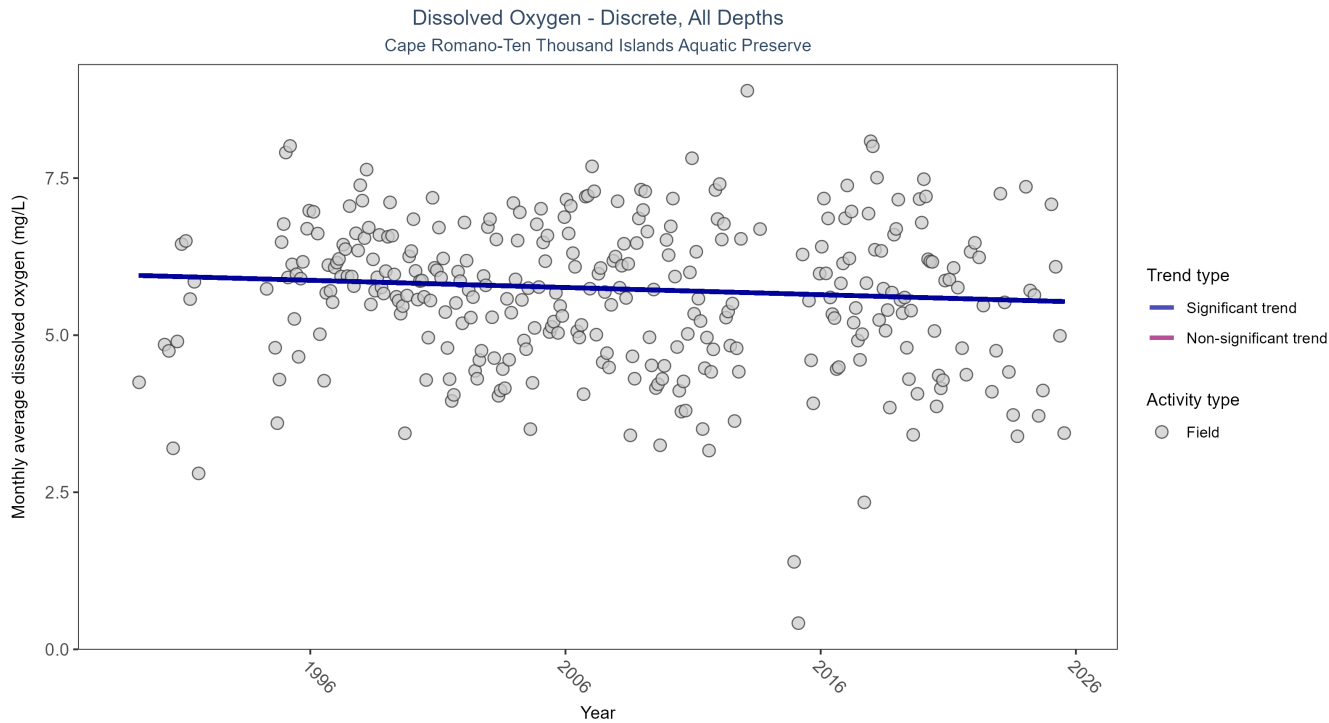


Figure 5: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Trend Analysis for Dissolved Oxygen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	10119	35	1989 - 2025	5.8	-0.077	5.9538	-0.0114	0.0498

Monthly average dissolved oxygen decreased by 0.01 mg/L per year.

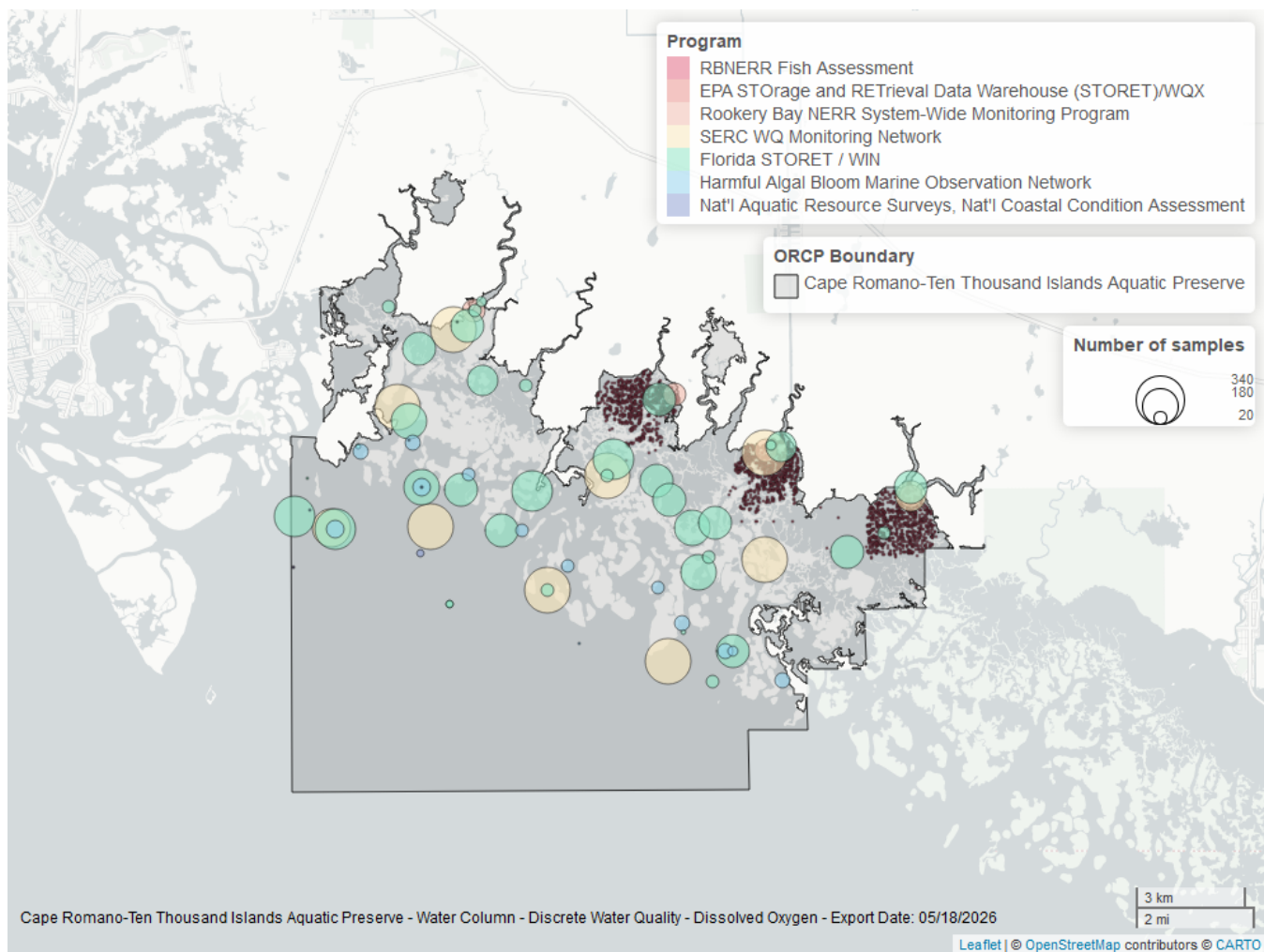


Figure 6: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 11: Programs contributing data for Dissolved Oxygen

ProgramID	N_Data	YearMin	YearMax
5002	4215	1989	2025
509	3024	1994	2008
4043	2478	1999	2020
95	390	1997	2018
354	325	2016	2023
118	10	2015	2021
103	2	2015	2015

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network³

4043 - RBNERR Fish Assessment⁷

95 - Harmful Algal Bloom Marine Observation Network⁸

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

pH - Discrete

Seasonal Kendall-Tau Trend Analysis

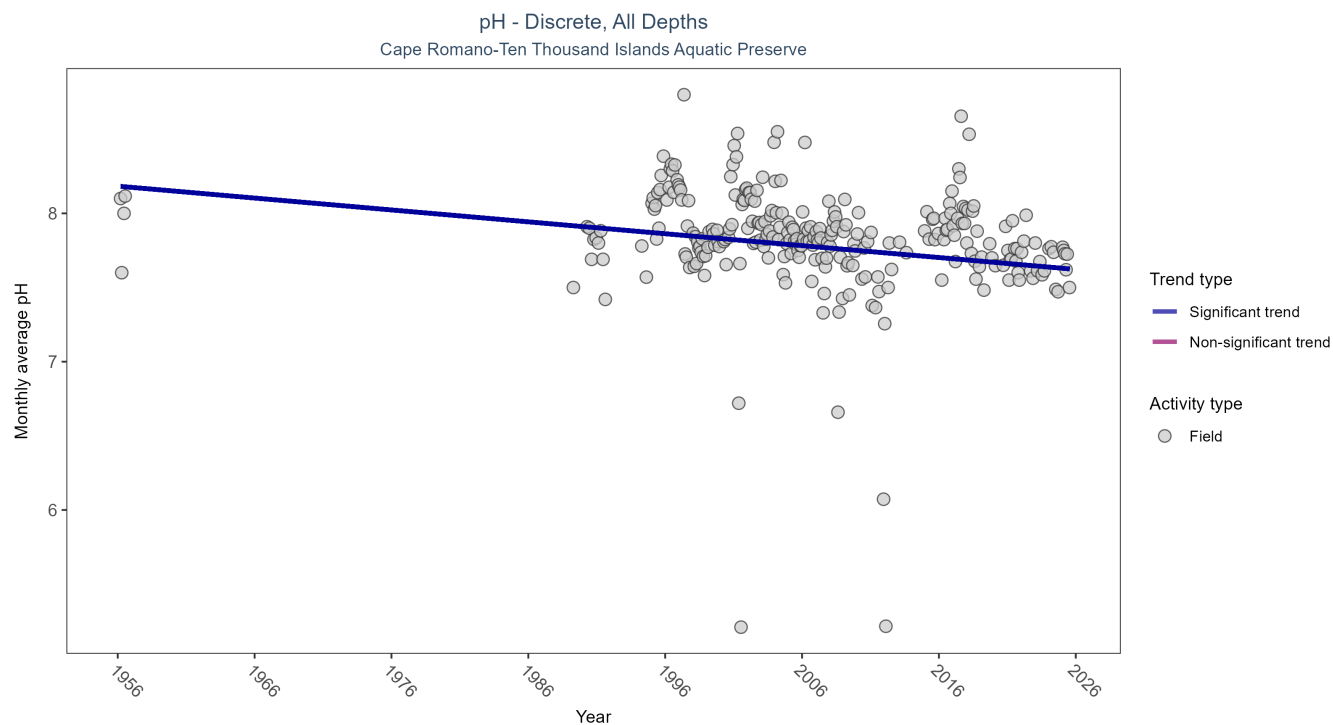


Figure 7: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Trend Analysis for pH

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	Decreasing trend	4301	36	1956 - 2025	7.9	-0.237	8.1834	-0.008	0

Monthly average pH decreased by 0.01 pH units per year.

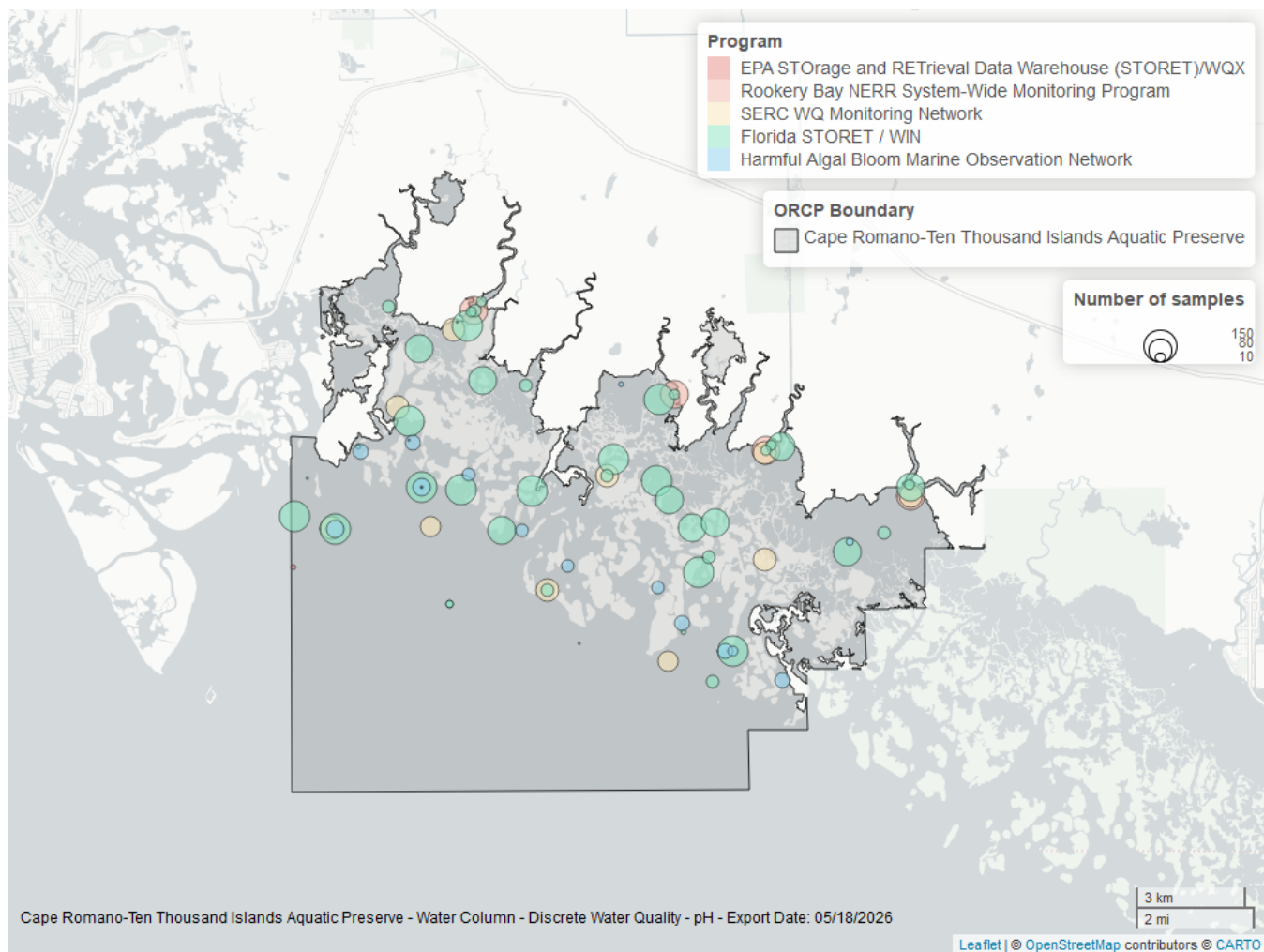


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 13: Programs contributing data for pH

ProgramID	N_Data	YearMin	YearMax
5002	3114	1989	2025
509	748	2001	2008
354	479	2016	2025
95	395	1956	2018
103	4	2015	2015

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network³

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

95 - Harmful Algal Bloom Marine Observation Network⁸

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Salinity - Discrete

Seasonal Kendall-Tau Trend Analysis

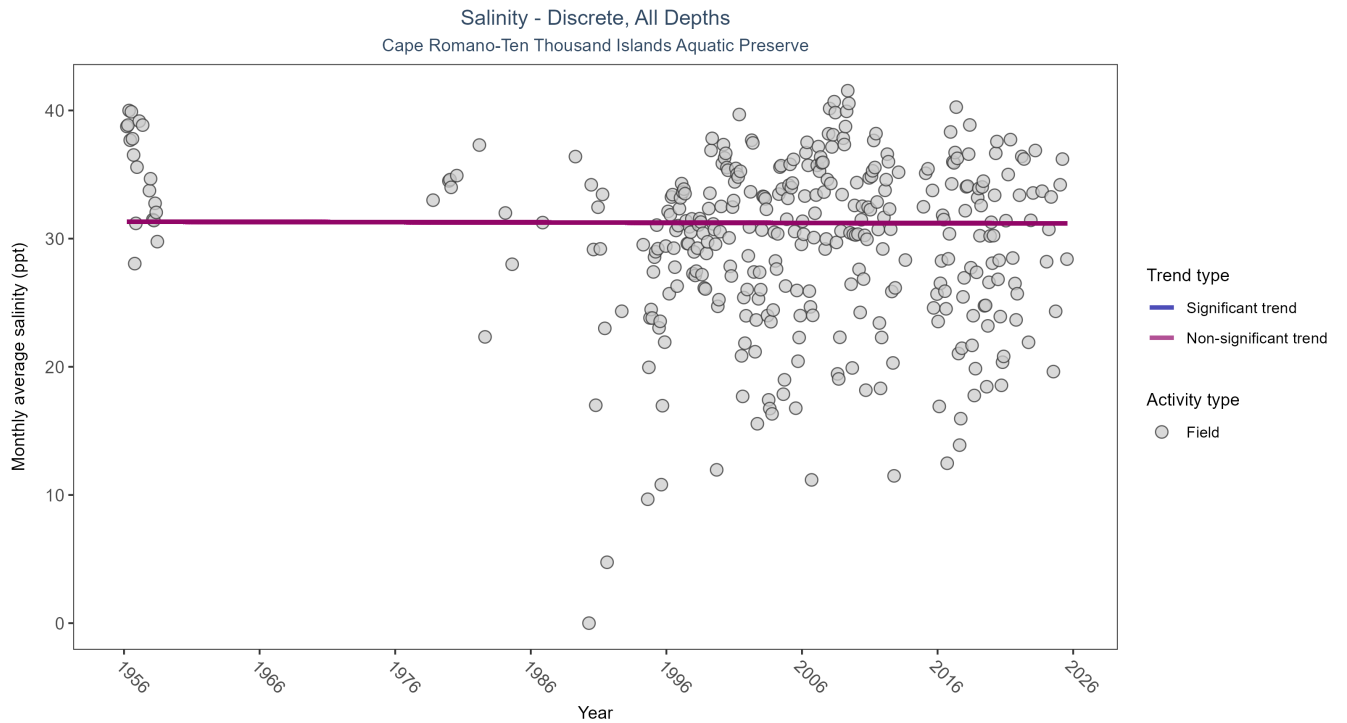


Figure 9: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 14: Seasonal Kendall-Tau Trend Analysis for Salinity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
All	No detectable trend	10478	45	1956 - 2025	32	-0.0049	31.3225	-0.002	0.94

Salinity showed no detectable trend between 1956 and 2025.

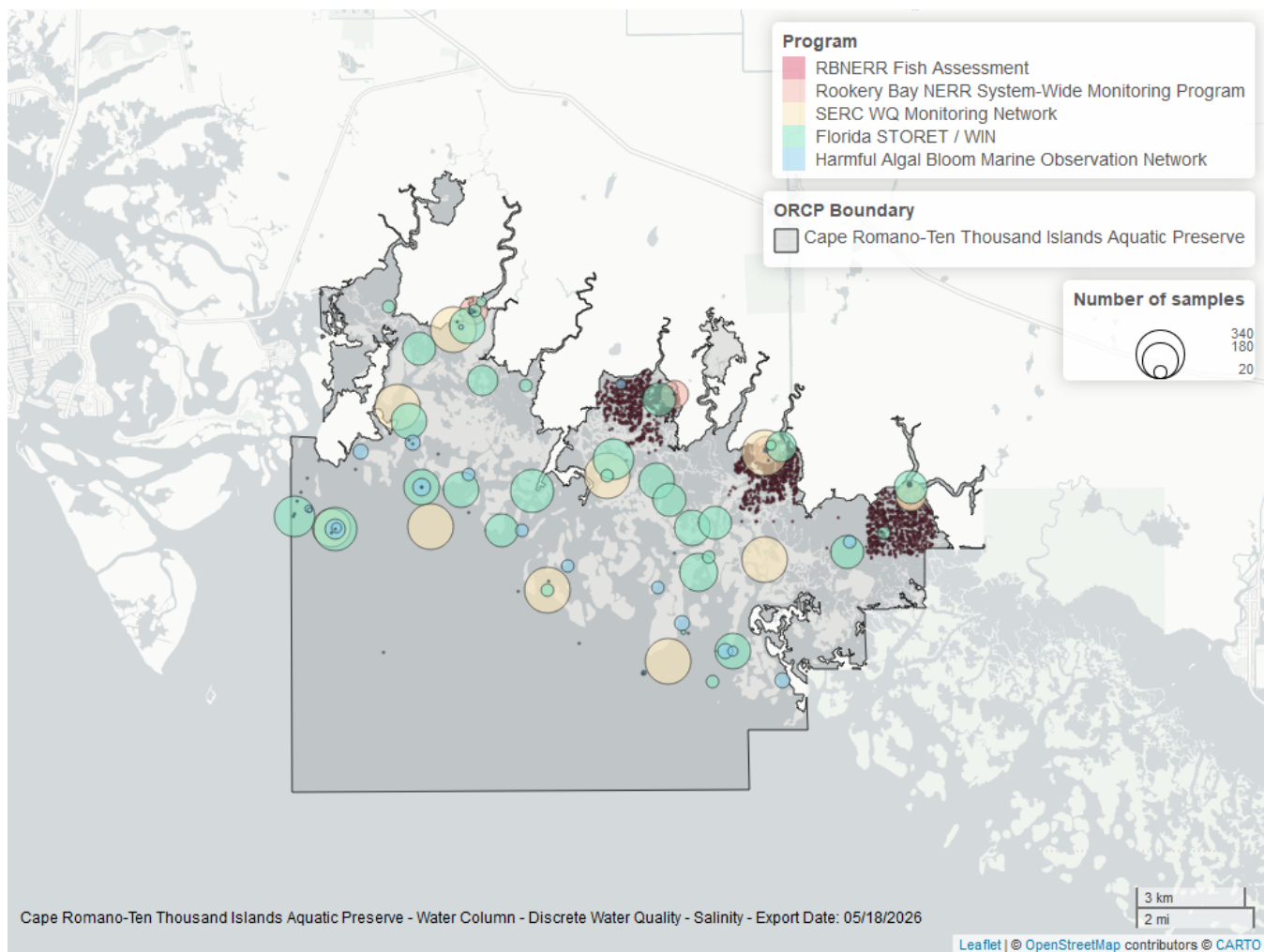


Figure 10: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 15: Programs contributing data for Salinity

ProgramID	N_Data	YearMin	YearMax
5002	4401	1989	2025
509	3026	1994	2008
4043	2538	1999	2020
95	532	1956	2018
354	521	2012	2025

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network³

4043 - RBNERR Fish Assessment⁷

95 - Harmful Algal Bloom Marine Observation Network⁸

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

Total Nitrogen - Discrete

Total Nitrogen Calculation:

The logic for calculated Total Nitrogen was provided by Kevin O'Donnell and colleagues at FDEP (with the help of Jay Silvanima, Watershed Monitoring Section). The following logic is used, in this order, based on the availability of specific nitrogen components.

- 1) $TN = TKN + NO3O2$;
- 2) $TN = TKN + NO3 + NO2$;
- 3) $TN = ORGN + NH4 + NO3O2$;
- 4) $TN = ORGN + NH4 + NO2 + NO3$;
- 5) $TN = TKN + NO3$;
- 6) $TN = ORGN + NH4 + NO3$;

Additional Information:

- Rules for use of sample fraction:
 - Florida Department of Environmental Protection (FDEP) report that if both “Total” and “Dissolved” components are reported, only “Total” is used. If the total is not reported, then the dissolved components are used as a best available replacement.
 - Total nitrogen calculations are done using nitrogen components with the same sample fraction, nitrogen components with mixed total/dissolved sample fractions are not used. In other words, total nitrogen can be calculated when TKN and NO3O2 are both total sample fractions, or when both are dissolved sample fractions. *Future calculations of total nitrogen values may be based on components with mixed sample fractions.*
- Values inserted into data:
 - ParameterName = “Total Nitrogen”
 - SEACAR_QAQCFlagCode = “1Q”
 - SEACAR_QAQC_Description = “SEACAR Calculated”

Seasonal Kendall-Tau Trend Analysis

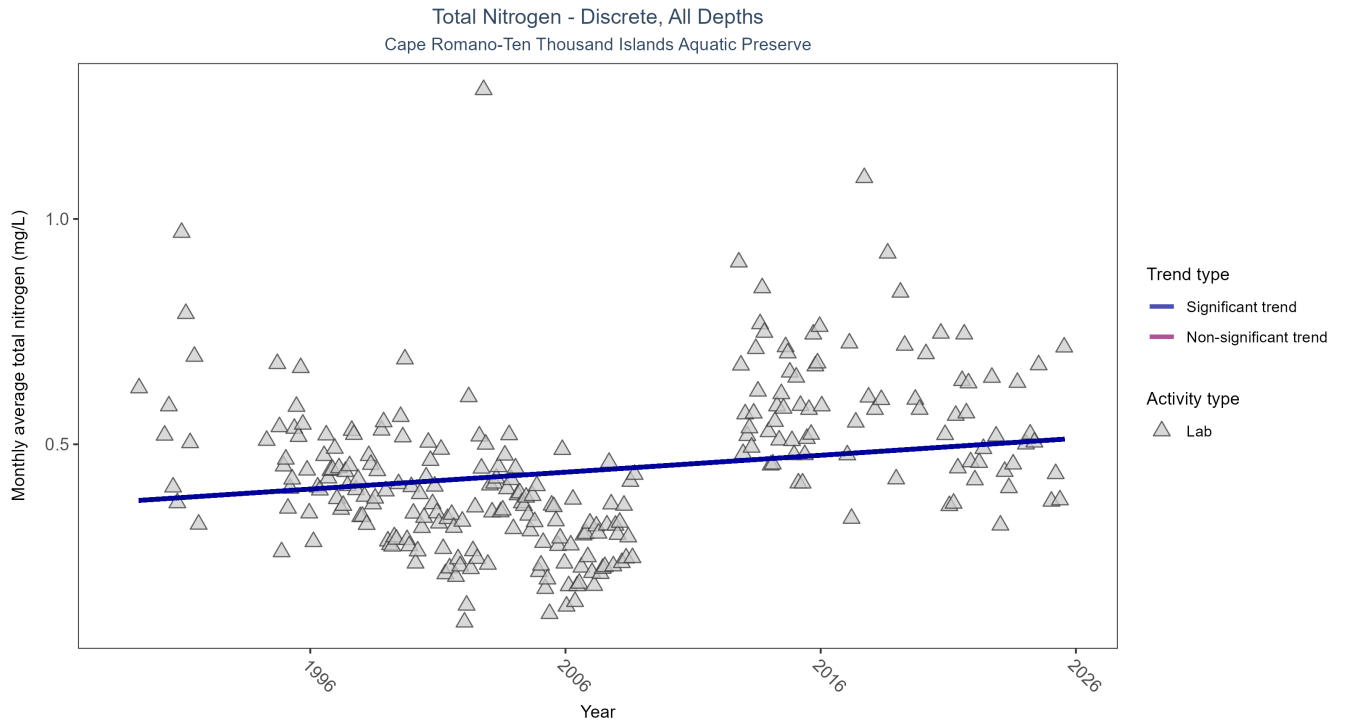


Figure 11: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Trend Analysis for Total Nitrogen

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	2023	32	1989 - 2025	0.3864	0.1315	0.3744	0.0037	0.0024

Monthly average total nitrogen increased by less than 0.01 mg/L per year.

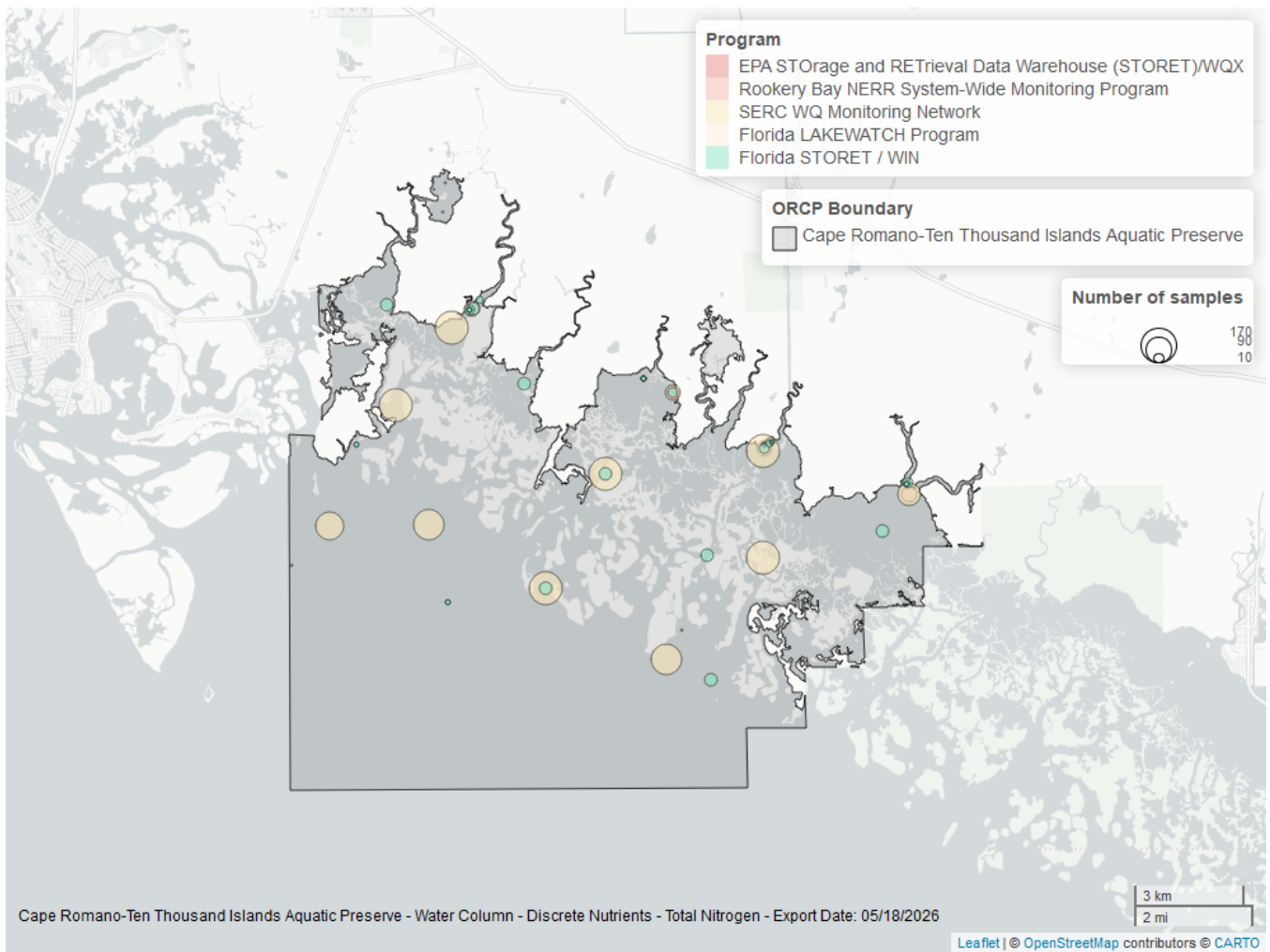


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 17: Programs contributing data for Total Nitrogen

ProgramID	N_Data	YearMin	YearMax
509	1512	1994	2008
5002	325	1989	2025
354	158	2012	2017
514	35	2001	2001
103	1	2015	2015

Program names:

509 - SERC Water Quality Monitoring Network³

5002 - Florida STORET / WIN²

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

514 - Florida LAKEWATCH Program⁴

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Total Phosphorus - Discrete

Seasonal Kendall-Tau Trend Analysis

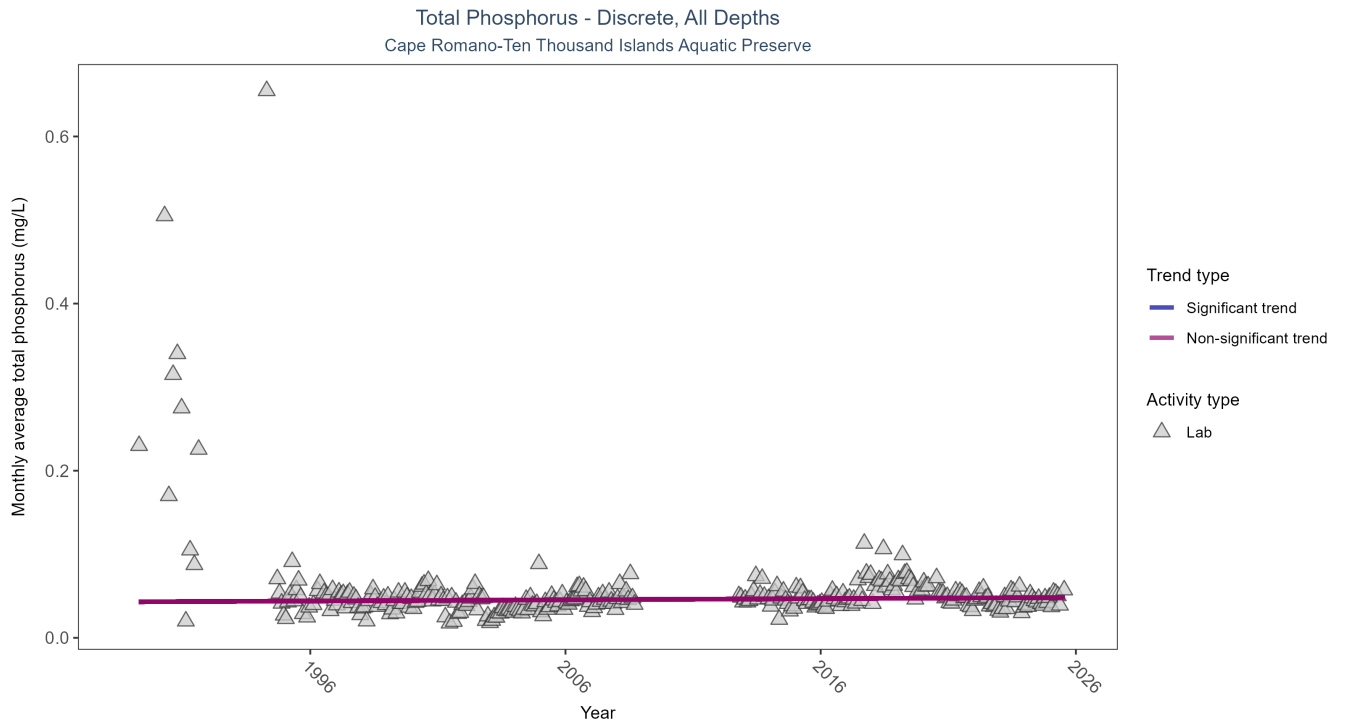


Figure 13: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 18: Seasonal Kendall-Tau Trend Analysis for Total Phosphorus

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	No detectable trend	2407	32	1989 - 2025	0.0435	0.0805	0.0431	0.0001	0.0505

Total phosphorus showed no detectable trend between 1989 and 2025.

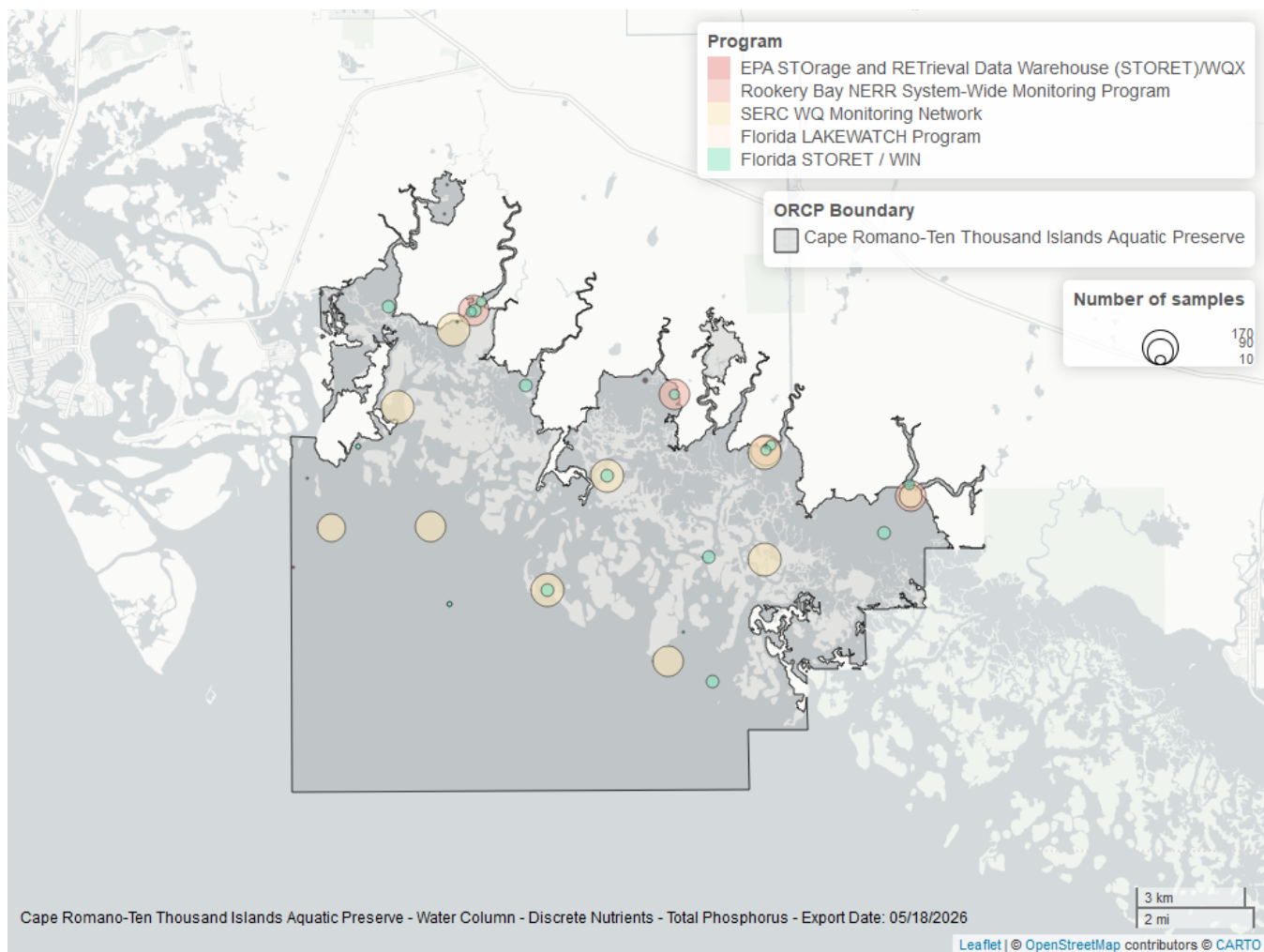


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 19: Programs contributing data for Total Phosphorus

ProgramID	N_Data	YearMin	YearMax
509	1499	1994	2008
354	589	2012	2025
5002	305	1989	2025
514	31	2001	2001
103	1	2015	2015

Program names:

509 - SERC Water Quality Monitoring Network³

[354](#) - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

5002 - Florida STORET / WIN²

514 - Florida LAKEWATCH Program⁴

103 - EPA STORage and RETrieval Data Warehouse (STORET)/WQX⁶

Turbidity - Discrete

Seasonal Kendall-Tau Trend Analysis

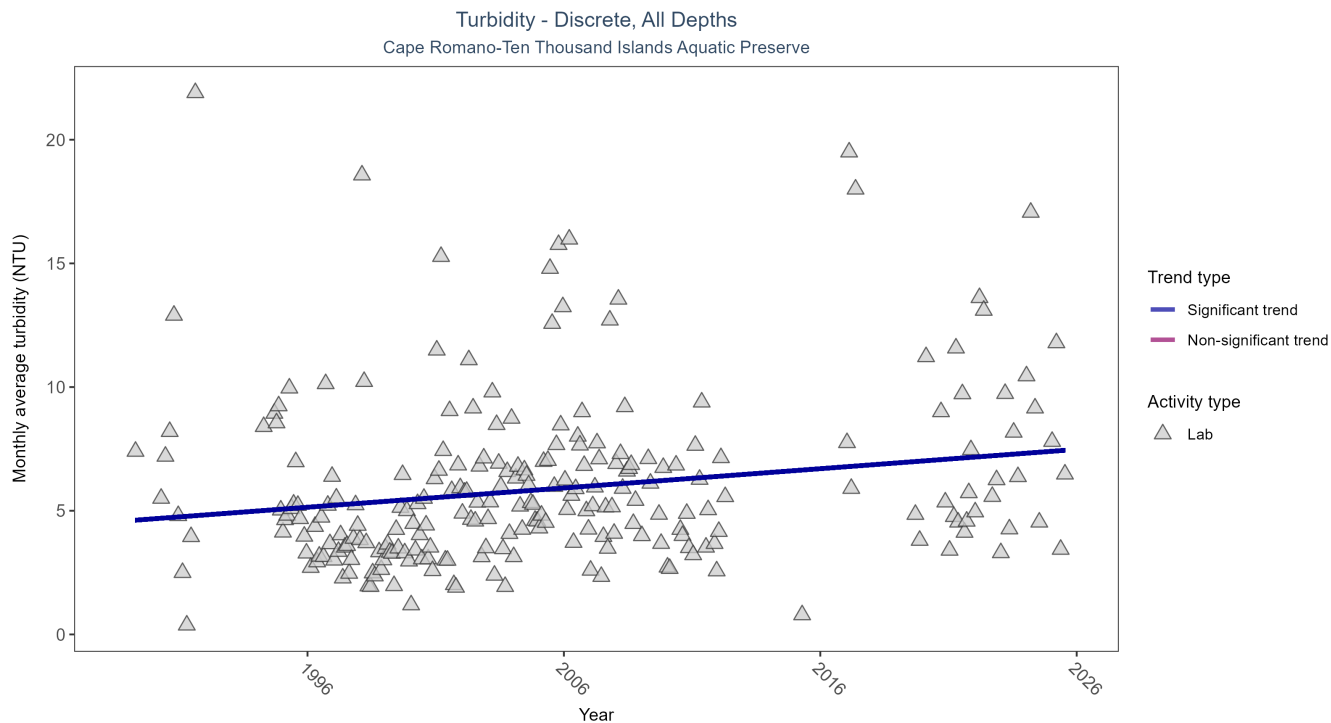


Figure 15: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 20: Seasonal Kendall-Tau Trend Analysis for Turbidity

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Lab	Increasing trend	4677	31	1989 - 2025	4	0.1565	4.5964	0.0779	0.0005

Monthly average turbidity increased by 0.08 NTU per year, indicating a decrease in water clarity.

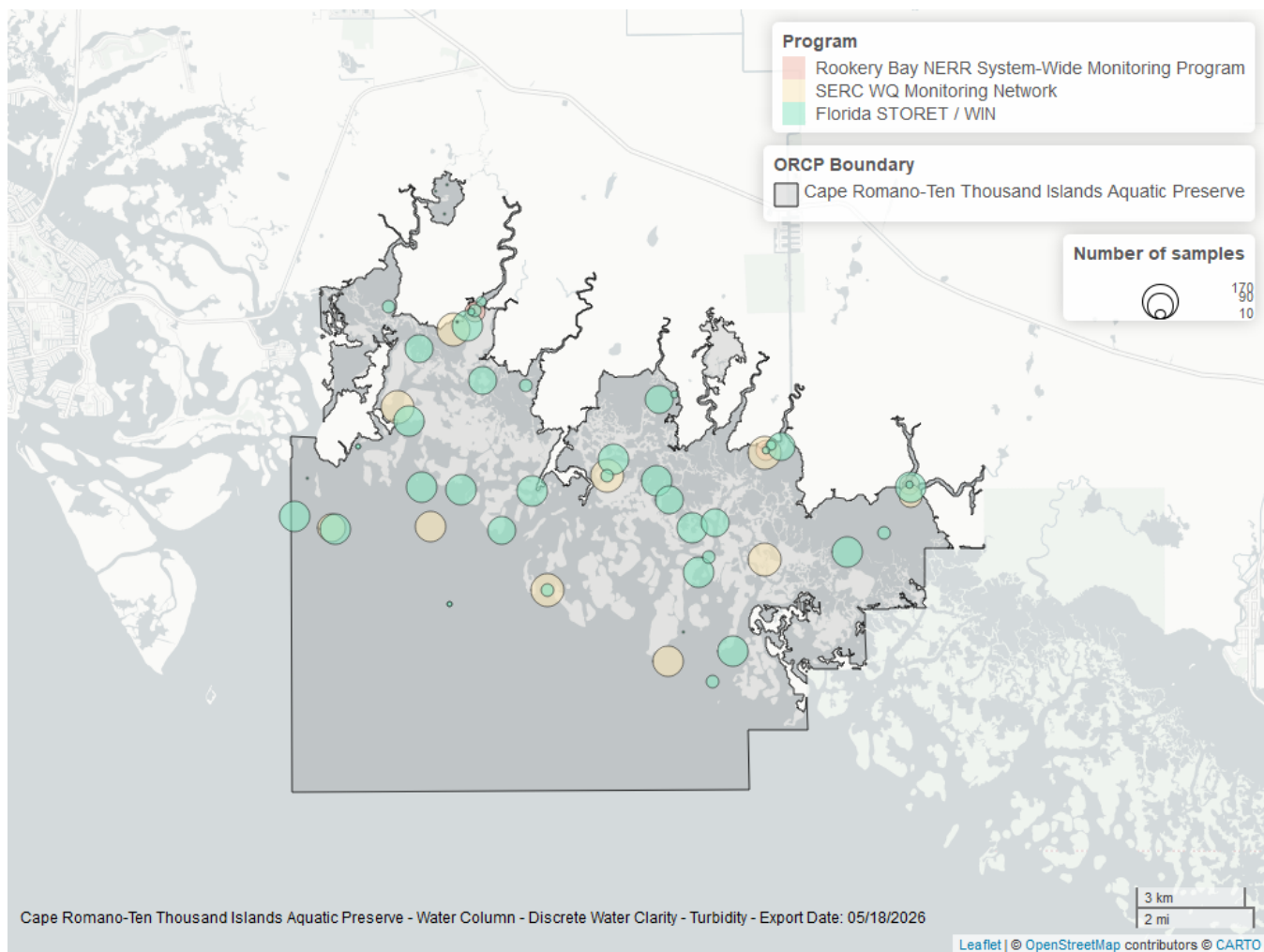


Figure 16: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 21: Programs contributing data for Turbidity

ProgramID	N_Data	YearMin	YearMax
5002	3174	1989	2025
509	1510	1994	2008
354	207	2002	2006

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network³

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

Water Temperature - Discrete

Seasonal Kendall-Tau Trend Analysis

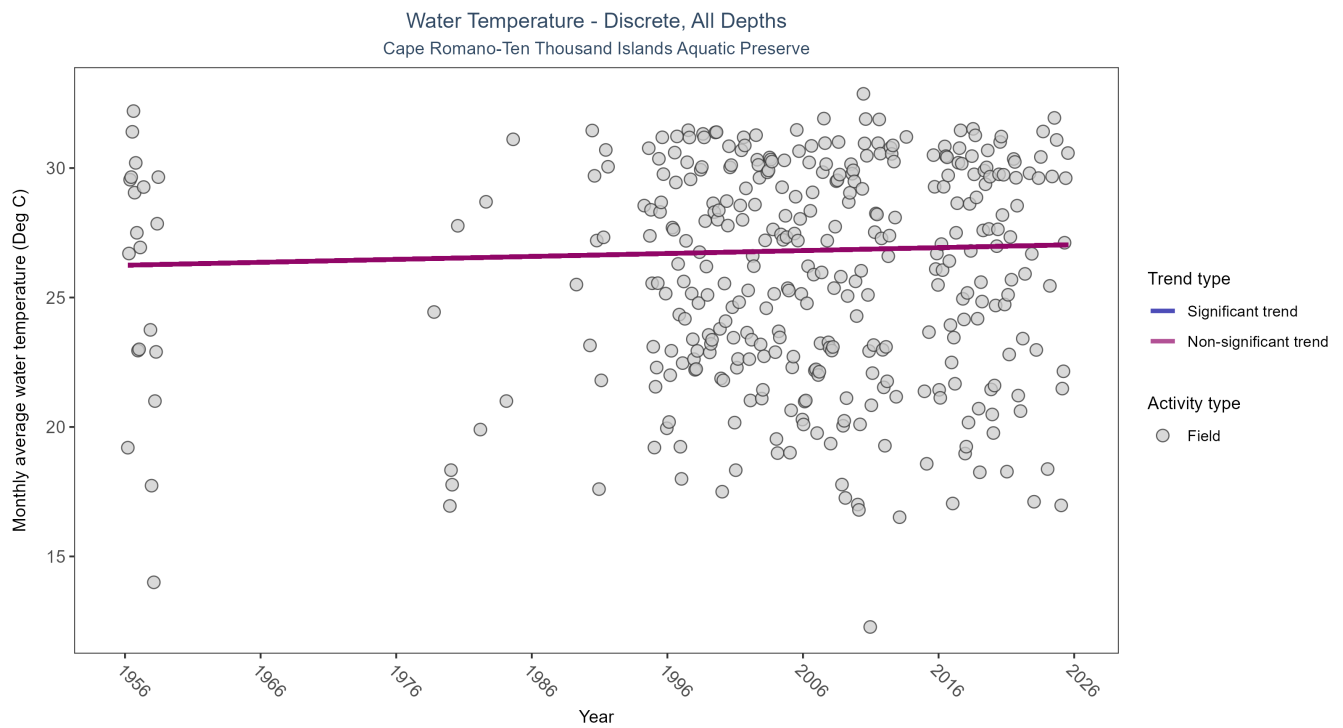


Figure 17: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 22: Seasonal Kendall-Tau Trend Analysis for Water Temperature

Activity Type	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
Field	No detectable trend	10694	43	1956 - 2025	26.8	0.0628	26.2463	0.0113	0.0788

Water temperature showed no detectable trend between 1956 and 2025.

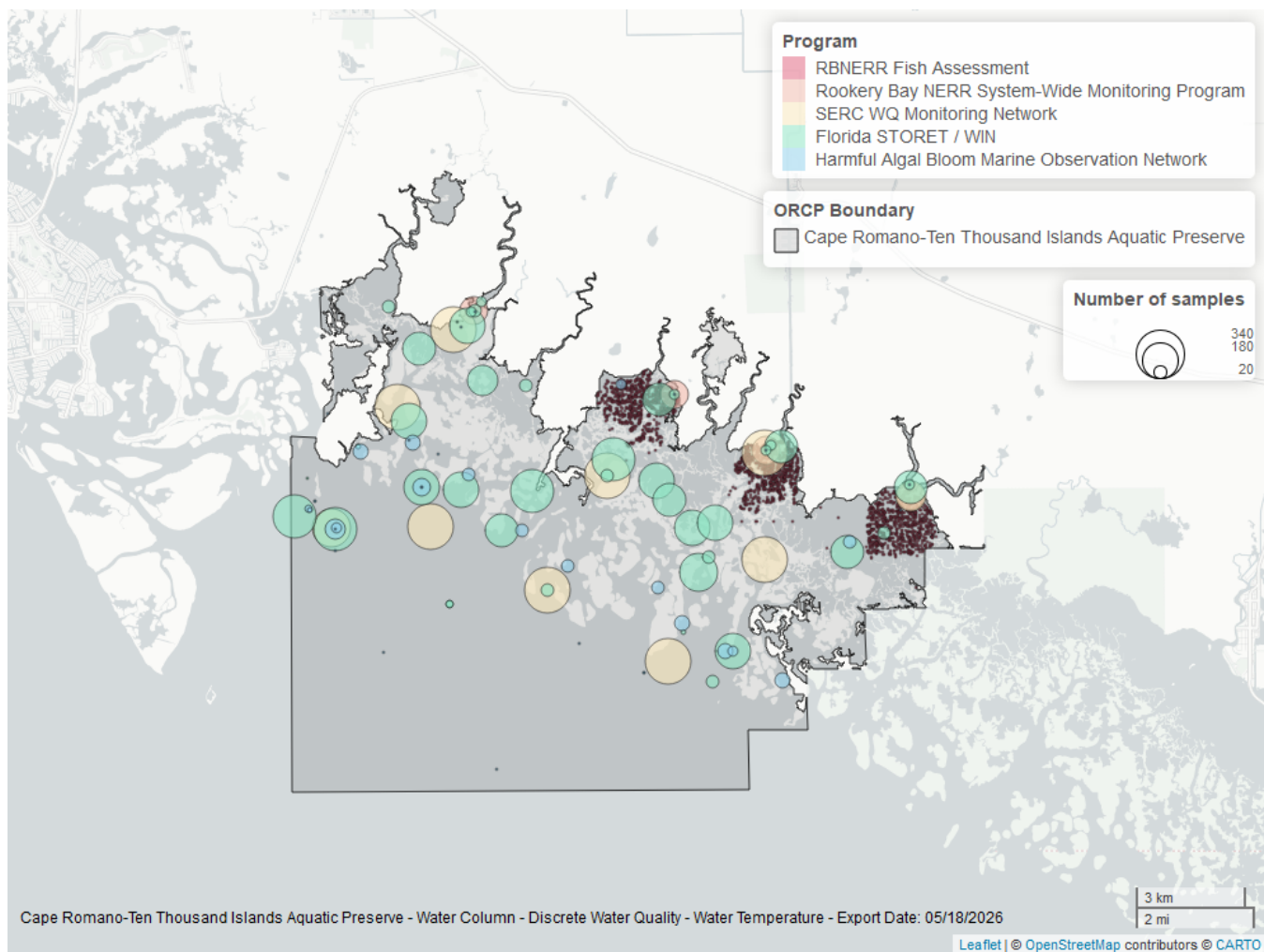


Figure 18: Map showing location of discrete water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Table 23: Programs contributing data for Water Temperature

ProgramID	N_Data	YearMin	YearMax
5002	4614	1989	2025
509	3026	1994	2008
4043	2532	1999	2020
354	489	2012	2025
95	482	1956	2018

Program names:

5002 - Florida STORET / WIN²

509 - SERC Water Quality Monitoring Network³

4043 - RBNERR Fish Assessment⁷

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

95 - Harmful Algal Bloom Marine Observation Network⁸

Water Quality - Continuous

The following files were used in the continuous analysis:

- *Combined_WQ_WC_NUT_cont_Chlorophyll_a_Uncorrected_for_Pheophytin-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Dissolved_Oxygen_Saturation-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Fluorescent_Dissolved_Organic_Matter-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_pH-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Salinity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Specific_Conductivity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Turbidity-2026-Mar-06.txt*
- *Combined_WQ_WC_NUT_cont_Water_Temperature-2026-Mar-06.txt*

Continuous monitoring locations in Cape Romano-Ten Thousand Islands Aquatic Preserve

Table 24: Station overview for Continuous parameters by Program

<i>ProgramID</i>	<i>ProgramLocationID</i>	<i>Years of Data</i>	<i>Use in Analysis</i>	<i>Parameters</i>
7	255123081321300	8	TRUE	Specific Conductivity
7	255123081321300	11	TRUE	Water Temperature, Salinity
7	255138081321701	10	TRUE	Water Temperature , Salinity , Specific Conductivity
7	255432081303900	17	TRUE	Specific Conductivity
7	255432081303900	21	TRUE	Water Temperature, Salinity
7	255443081314700	5	FALSE	Water Temperature, Salinity
7	255532081314300	3	FALSE	Water Temperature, Salinity
7	255534081324000	17	TRUE	Specific Conductivity
7	255534081324000	20	TRUE	Water Temperature, Salinity
7	255654081350200	17	TRUE	Specific Conductivity
7	255654081350200	20	TRUE	Water Temperature, Salinity
7	255732081363700	5	TRUE	Water Temperature, Salinity
354	rkbfbwq	25	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkbfiwq	25	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkbmbwq	27	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity
354	rkbpbwq	11	TRUE	pH , Dissolved Oxygen Saturation, Dissolved Oxygen , Turbidity , Water Temperature , Salinity , Specific Conductivity

Program names:

7 - National Water Information System⁹

354 - Rookery Bay National Estuarine Research Reserve System-Wide Monitoring Program¹

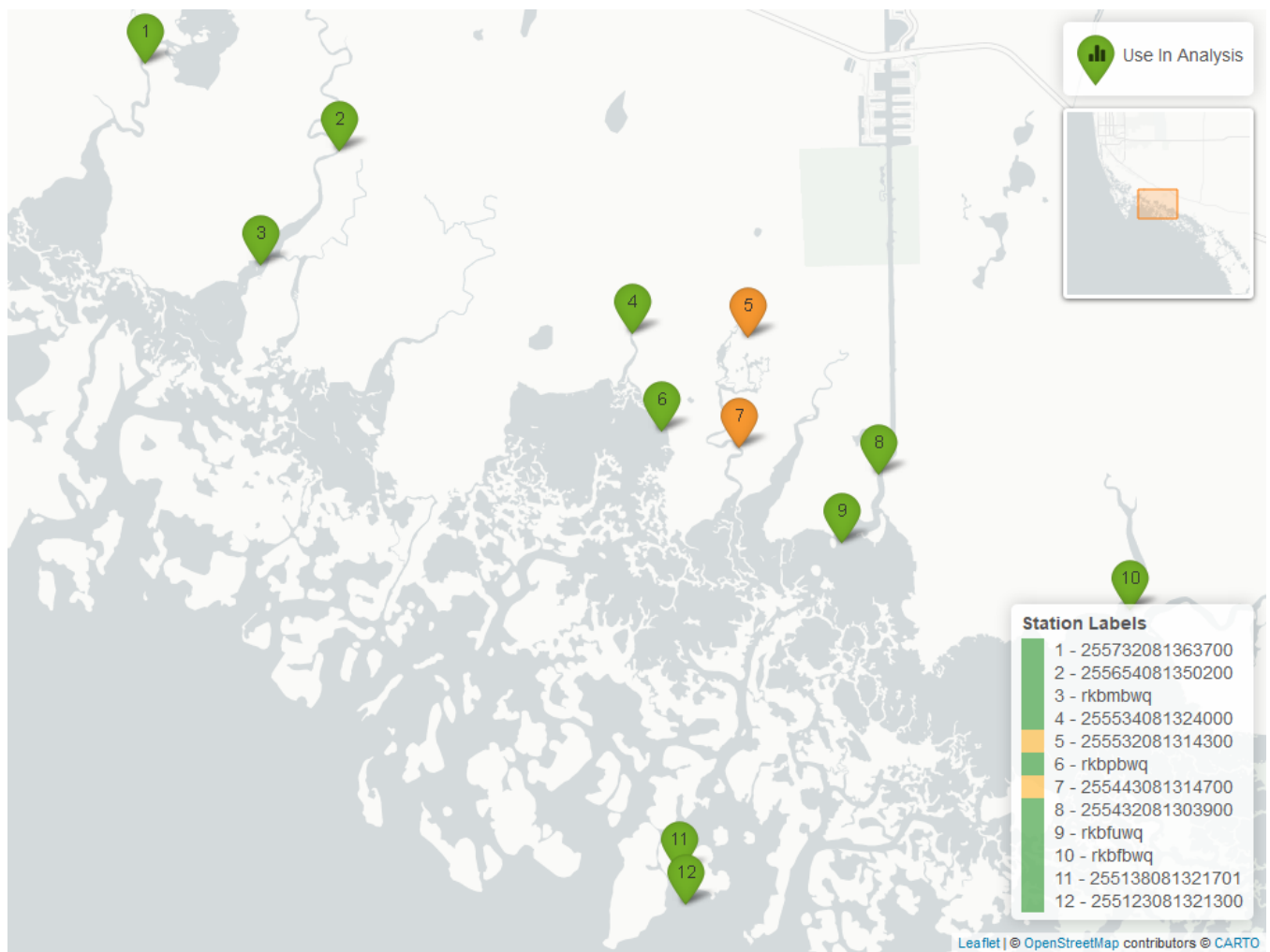


Figure 19: Map showing continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. Sites marked as *Use In Analysis* (green) are featured in this report.

pH - Continuous

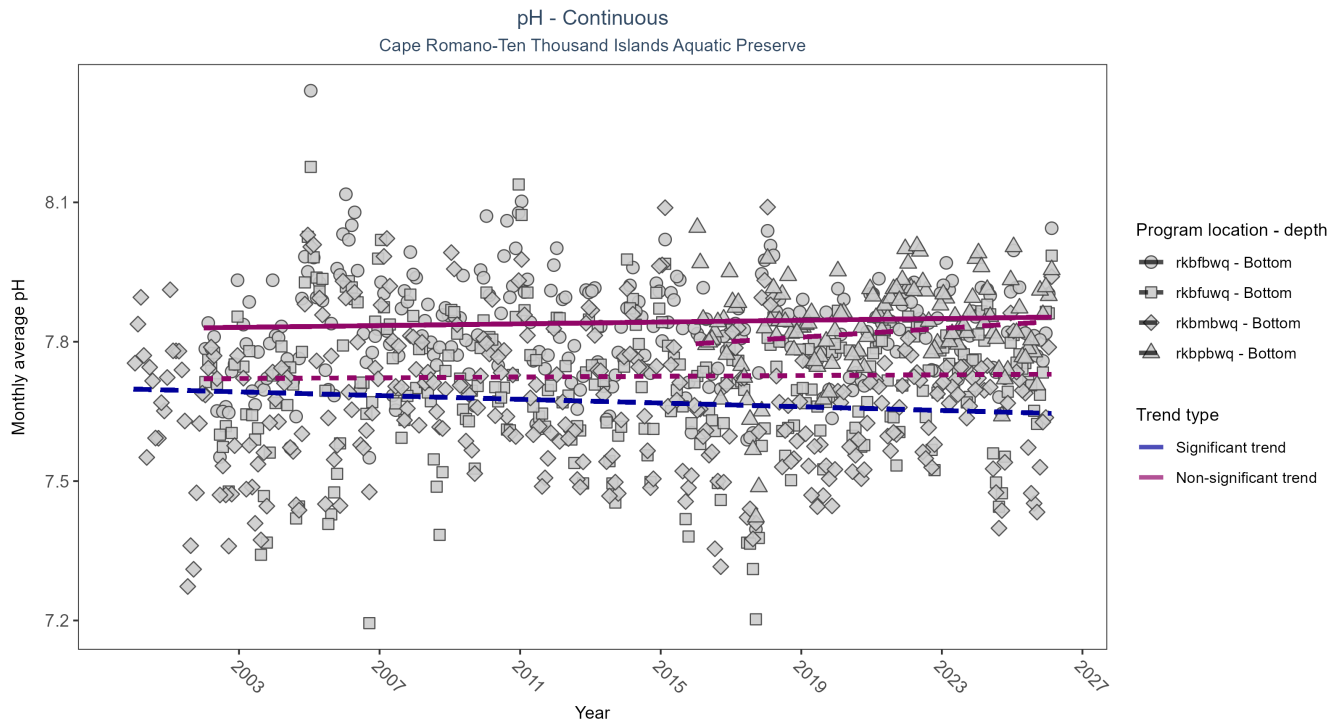


Figure 20: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 25: Seasonal Kendall-Tau Results for pH - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwbq	No significant trend	670684	25	2002 - 2026	7.8	0.05	7.83	0	0.21
rkbfuwq	No significant trend	689115	25	2002 - 2026	7.7	0.02	7.72	0	0.61
rkmbwbq	Significantly decreasing trend	727383	27	2000 - 2026	7.7	-0.10	7.70	0	0.02
rkbpbwq	No significant trend	328113	11	2016 - 2026	7.8	0.13	7.80	0	0.08

At one program location, monthly average pH decreased by less than 0.01 pH units per year. No detectable change in monthly average pH was observed at three locations.

Dissolved Oxygen Saturation - Continuous

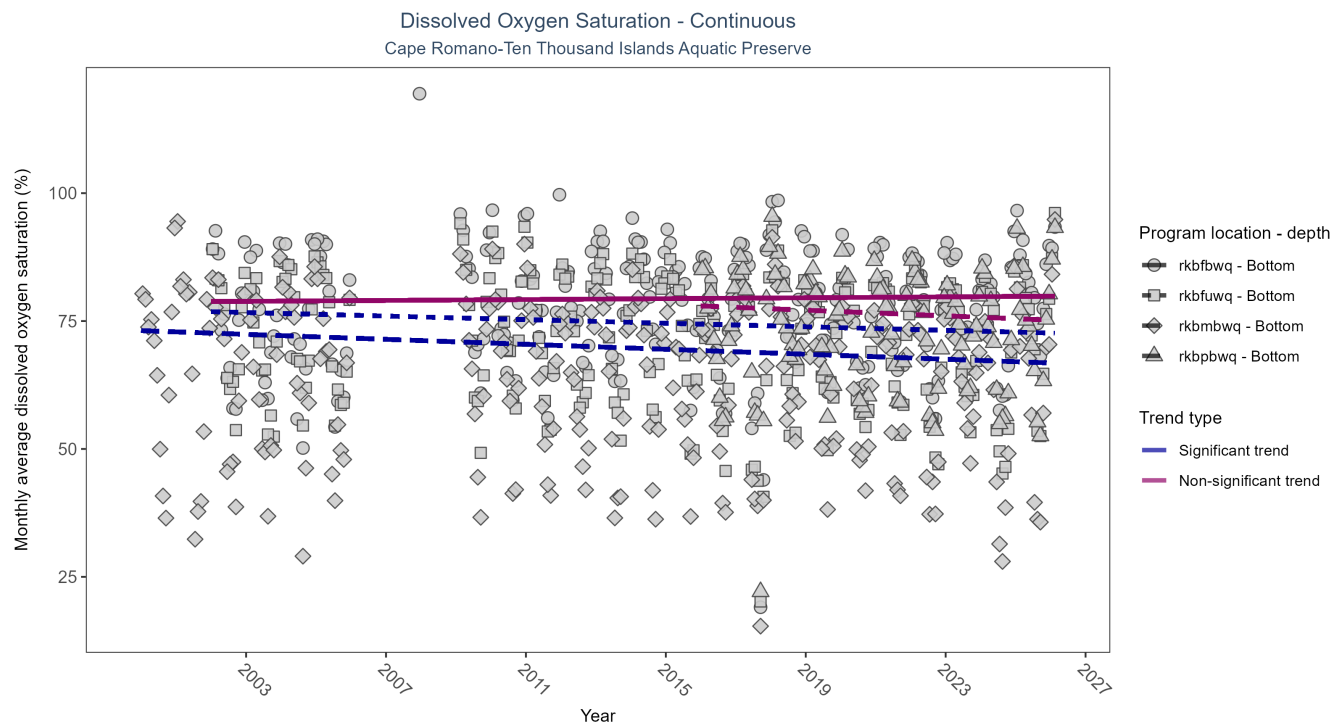


Figure 21: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 26: Seasonal Kendall-Tau Results for Dissolved Oxygen Saturation - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwbq	No significant trend	608215	23	2002 - 2026	79.0	0.04	78.81	0.04	0.43
rkbfuwq	Significantly decreasing trend	636774	22	2002 - 2026	72.6	-0.20	76.84	-0.17	0.00
rkmbbwq	Significantly decreasing trend	663443	24	2000 - 2026	65.3	-0.21	73.13	-0.24	0.00
rkbpbwq	No significant trend	336249	11	2016 - 2026	72.7	-0.13	77.96	-0.28	0.08

At two program locations, monthly average dissolved oxygen saturation decreased by 0.17% per year at one site and by 0.24% per year at the other. No detectable change in monthly average dissolved oxygen saturation was observed at two locations.

Dissolved Oxygen - Continuous

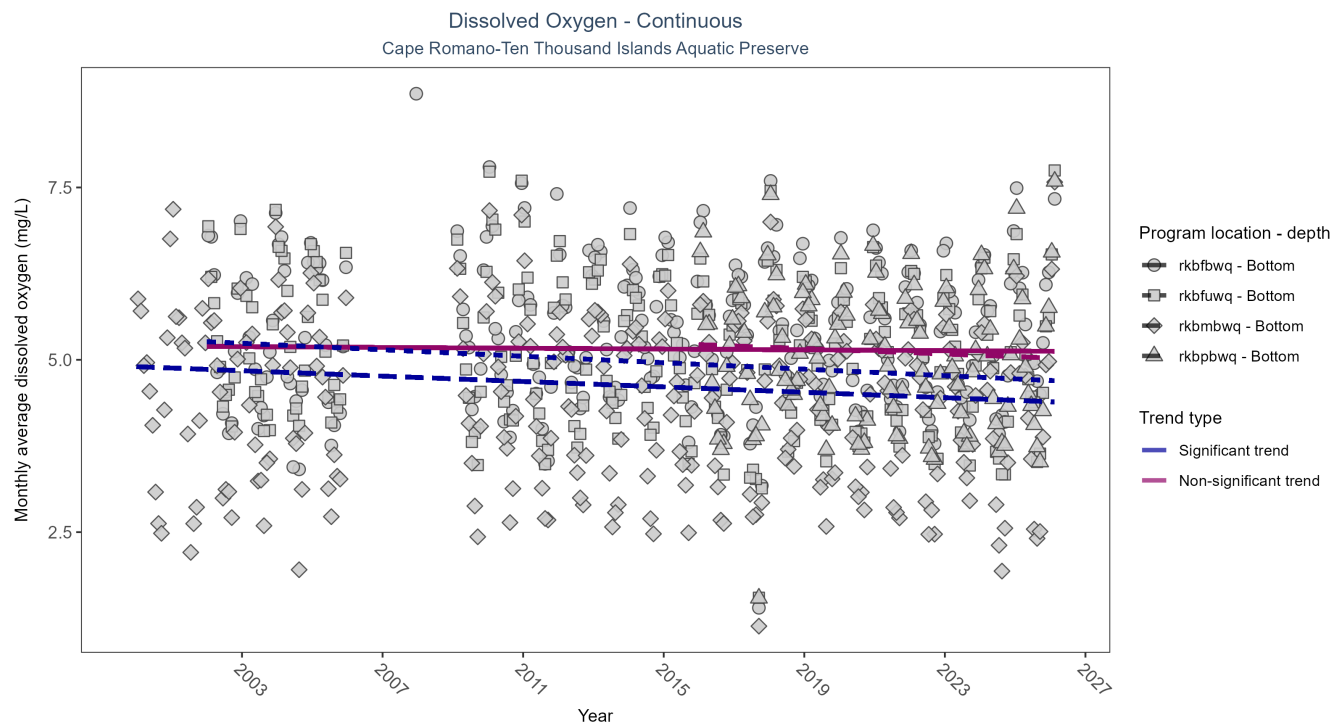


Figure 22: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 27: Seasonal Kendall-Tau Results for Dissolved Oxygen - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwbq	No significant trend	603909	23	2002 - 2026	5.4	-0.02	5.19	0.00	0.62
rkbfuwq	Significantly decreasing trend	636515	22	2002 - 2026	5.1	-0.29	5.26	-0.02	0.00
rkmbwbq	Significantly decreasing trend	657119	24	2000 - 2026	4.4	-0.24	4.90	-0.02	0.00
rkbpbwq	No significant trend	332812	11	2016 - 2026	4.9	-0.14	5.22	-0.02	0.08

At two program locations, monthly average dissolved oxygen decreased by 0.02 mg/L per year. No detectable change in monthly average dissolved oxygen was observed at two locations.

Turbidity - Continuous

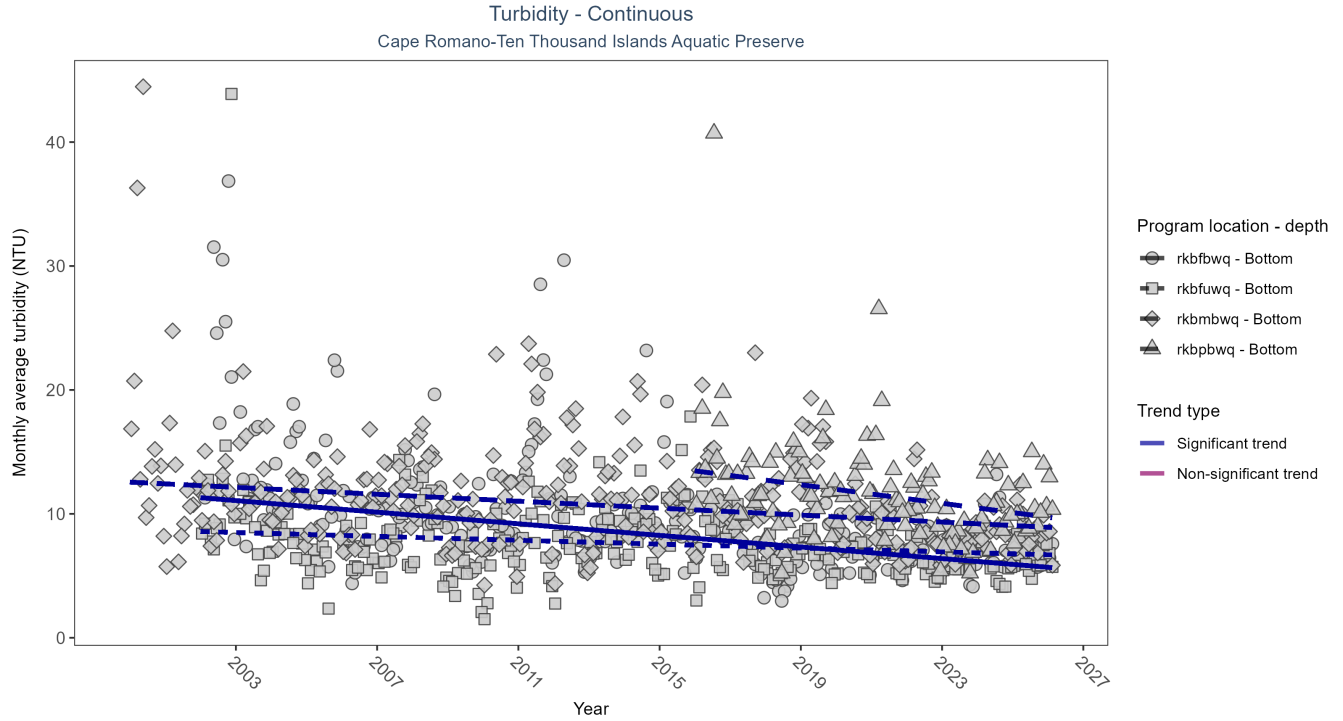


Figure 23: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 28: Seasonal Kendall-Tau Results for Turbidity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwq	Significantly decreasing trend	682915	25	2002 - 2026	7	-0.40	11.30	-0.23	0
rkbfuwq	Significantly decreasing trend	657778	25	2002 - 2026	6	-0.19	8.57	-0.08	0
rkmbwq	Significantly decreasing trend	714237	27	2000 - 2026	9	-0.24	12.56	-0.14	0
rkbpbwq	Significantly decreasing trend	331299	11	2016 - 2026	9	-0.31	13.47	-0.37	0

At four program locations, monthly average turbidity decreased between 0.08 and 0.37 NTU per year.

Water Temperature - Continuous - Program 7

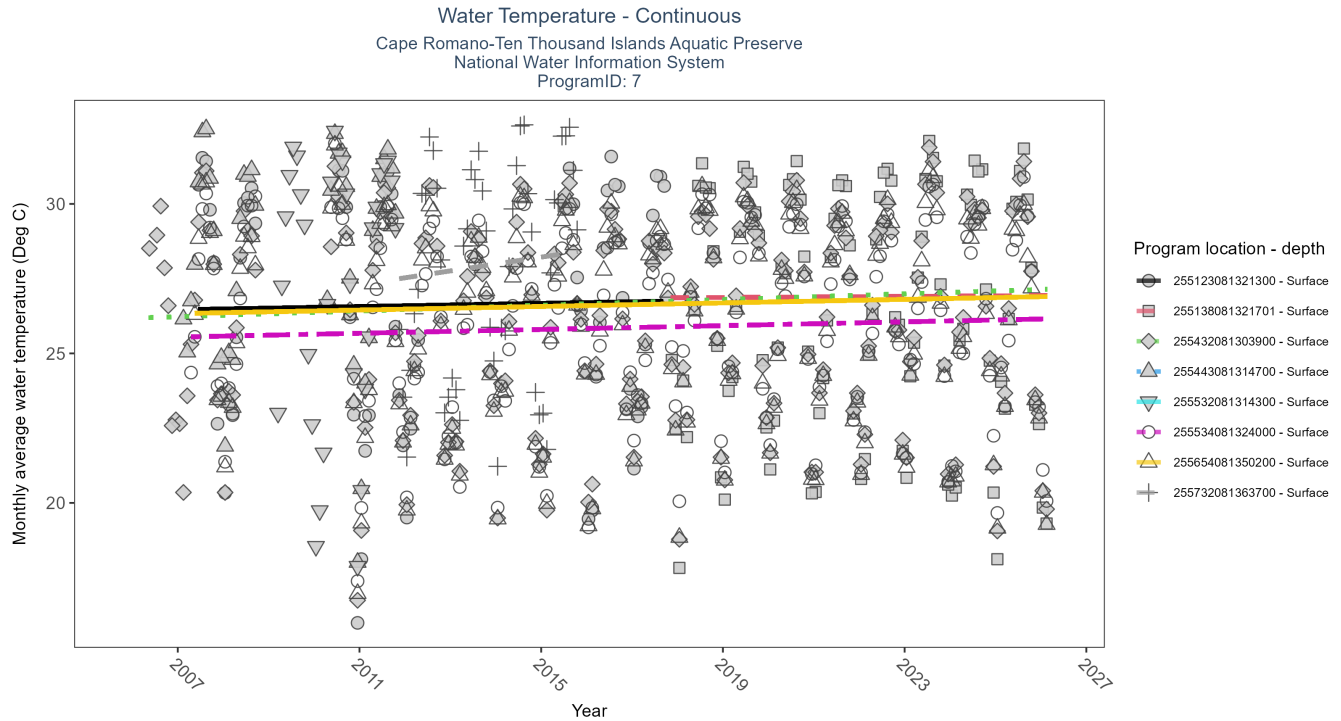


Figure 24: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 29: Seasonal Kendall-Tau Results for Water Temperature - Program 7

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1818	8	2007 - 2017	27.3	0.1	26.46	0.03	0.38
255138081321701	No significant trend	3005	10	2017 - 2026	27.0	0.02	26.85	0.01	0.92
255432081303900	Significantly increasing trend	6527	20	2006 - 2026	27.0	0.17	26.18	0.05	0
255443081314700	Insufficient data to calculate trend	2011	4	2007 - 2011	29.3	-	-	-	-
255532081314300	Insufficient data to calculate trend	906	3	2009 - 2011	29.3	-	-	-	-
255534081324000	Significantly increasing trend	6225	19	2007 - 2026	26.6	0.14	25.55	0.03	0.01
255654081350200	Significantly increasing trend	6222	19	2007 - 2026	26.7	0.11	26.33	0.03	0.03
255732081363700	No significant trend	1435	5	2011 - 2015	28.4	0.24	27.3	0.23	0.11

At six program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at four locations. There was insufficient data to fit a model for two locations.

Water Temperature - Continuous - Program 354

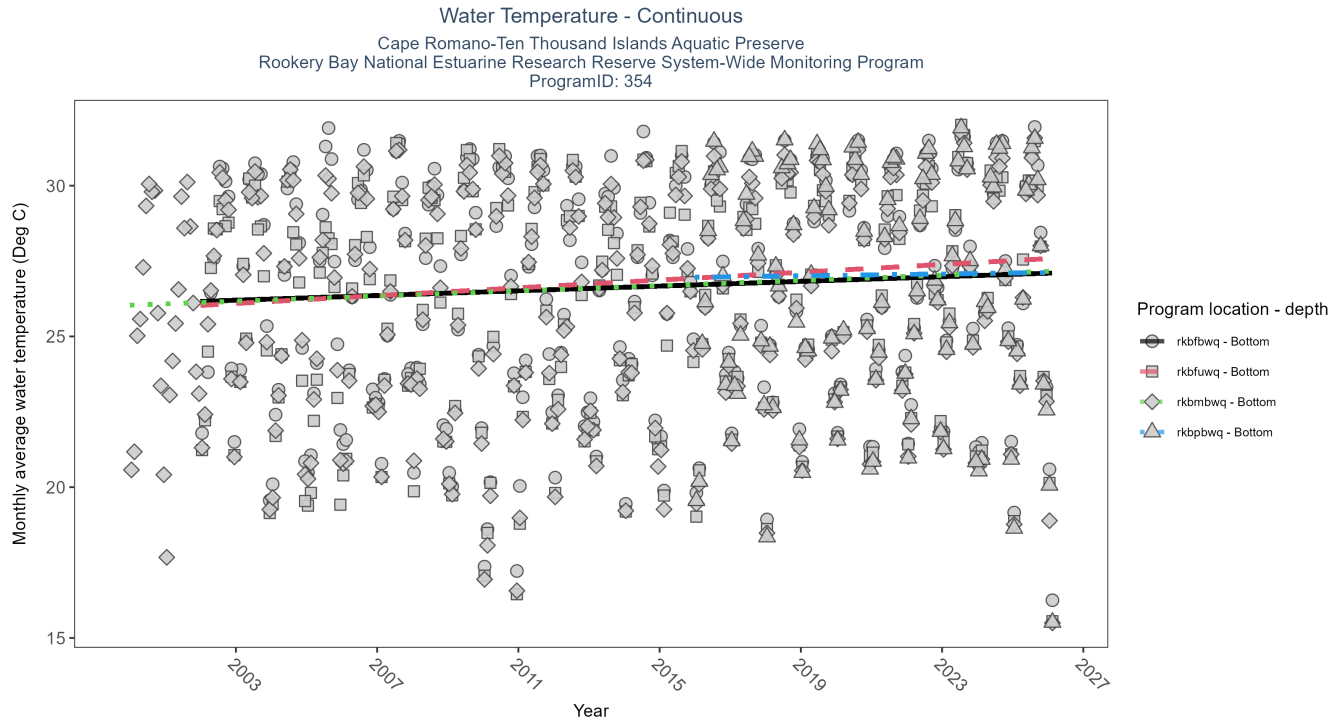


Figure 25: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 30: Seasonal Kendall-Tau Results for Water Temperature - Program 354

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfbwq	Significantly increasing trend	719021	25	2002 - 2026	27.0	0.22	26.16	0.04	0.00
rkbfuwq	Significantly increasing trend	727501	25	2002 - 2026	26.9	0.26	26.03	0.06	0.00
rkmbbwq	Significantly increasing trend	762033	27	2000 - 2026	26.8	0.25	26.04	0.04	0.00
rkbpbwq	No significant trend	337833	11	2016 - 2026	27.4	0.05	26.96	0.02	0.55

At six program locations, monthly average water temperature increased between 0.03 and 0.06°C per year. No detectable change in monthly average water temperature was observed at four locations. There was insufficient data to fit a model for two locations.

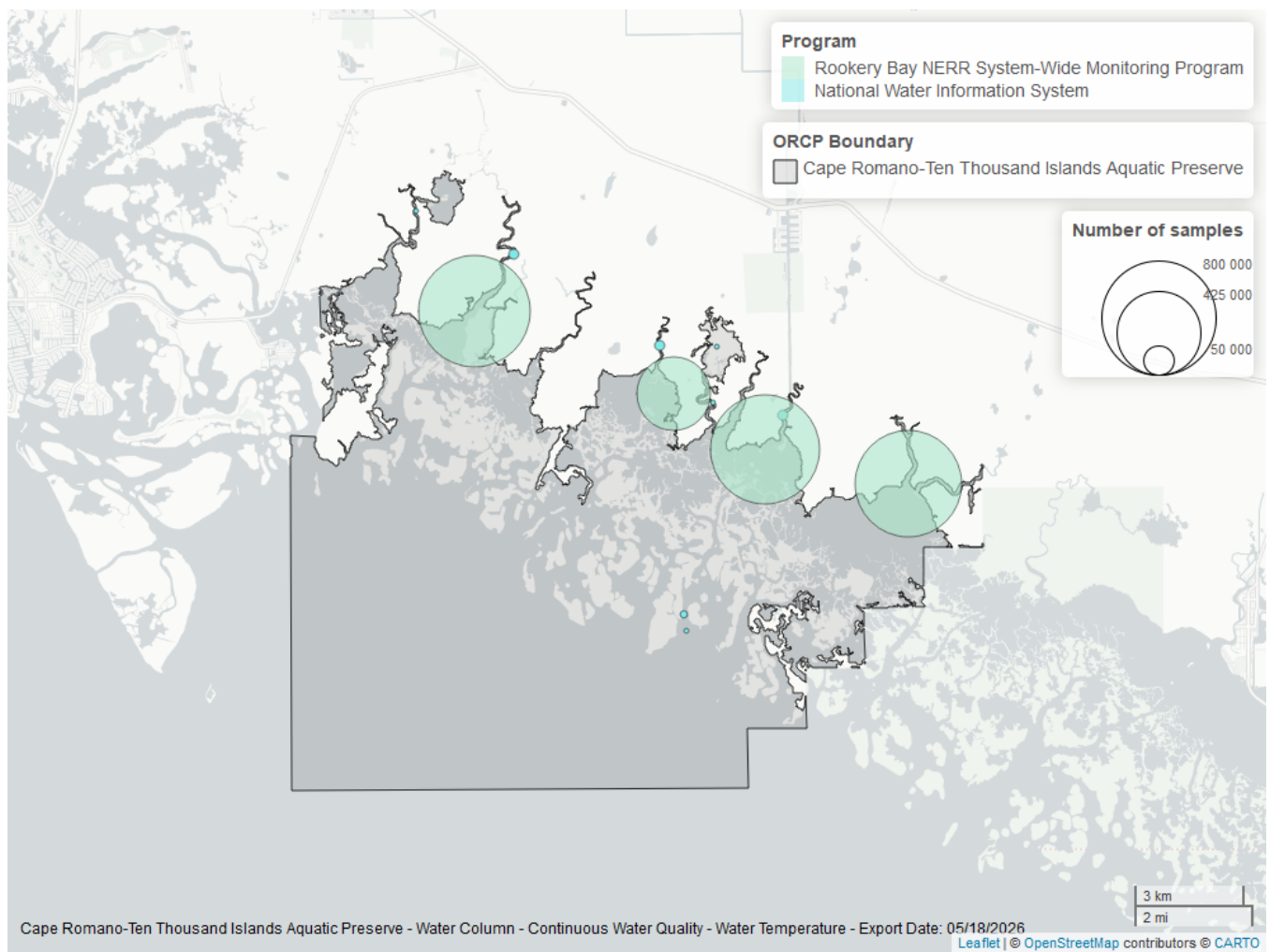


Figure 26: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Salinity - Continuous - Program 7

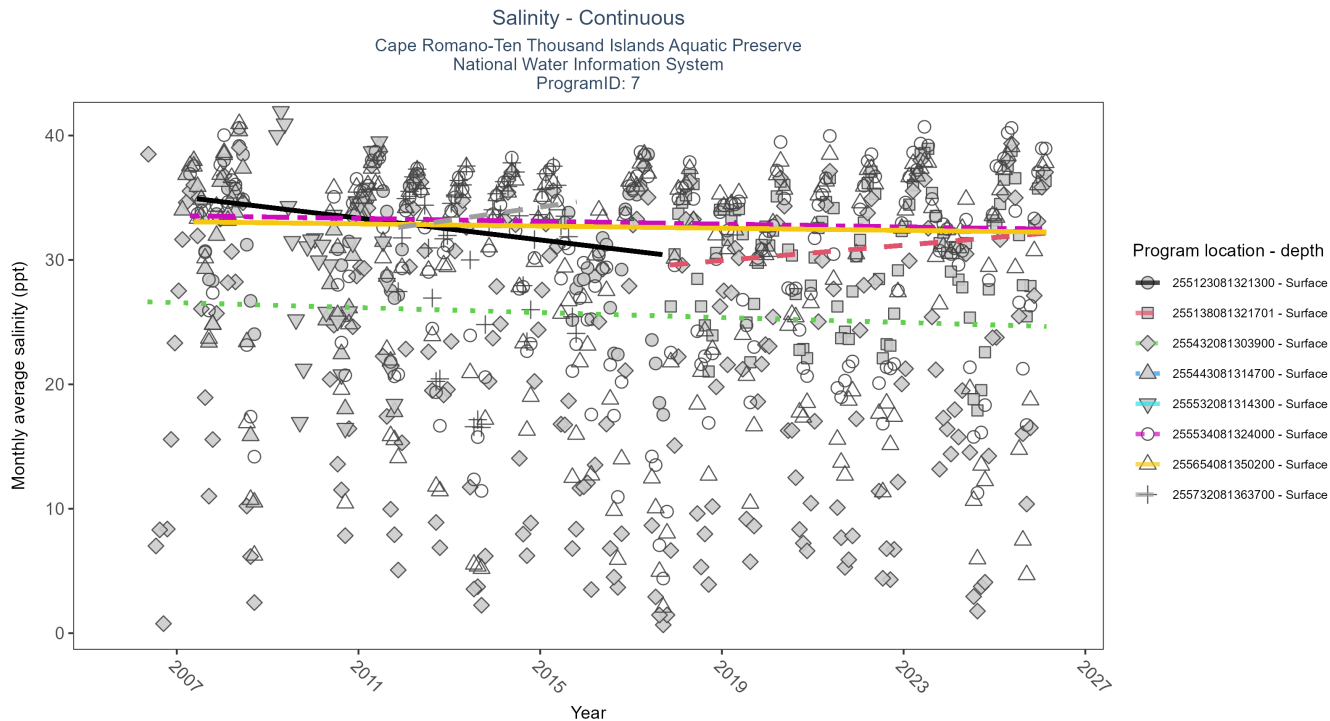


Figure 27: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 31: Seasonal Kendall-Tau Results for Salinity - Program 7

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	Significantly decreasing trend	1809	8	2007 - 2017	32	-0.23	35.11	-0.44	0.02
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	32	0.17	29.33	0.31	0.03
255432081303900	No significant trend	6468	20	2006 - 2026	22	-0.08	26.67	-0.1	0.11
255443081314700	Insufficient data to calculate trend	1465	4	2007 - 2011	32	-	-	-	-
255532081314300	Insufficient data to calculate trend	902	3	2009 - 2011	31	-	-	-	-
255534081324000	No significant trend	6162	19	2007 - 2026	32	-0.04	33.56	-0.06	0.36
255654081350200	No significant trend	6183	19	2007 - 2026	32	-0.04	33.07	-0.04	0.36
255732081363700	No significant trend	1434	5	2011 - 2015	34	0.25	32.15	0.52	0.1

At one program location, monthly average salinity increased by 0.31 ppt per year. At two program locations, monthly average salinity decreased by 0.06 ppt per year at one site and by 0.44 ppt per year at the other. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for two locations.

Salinity - Continuous - Program 354

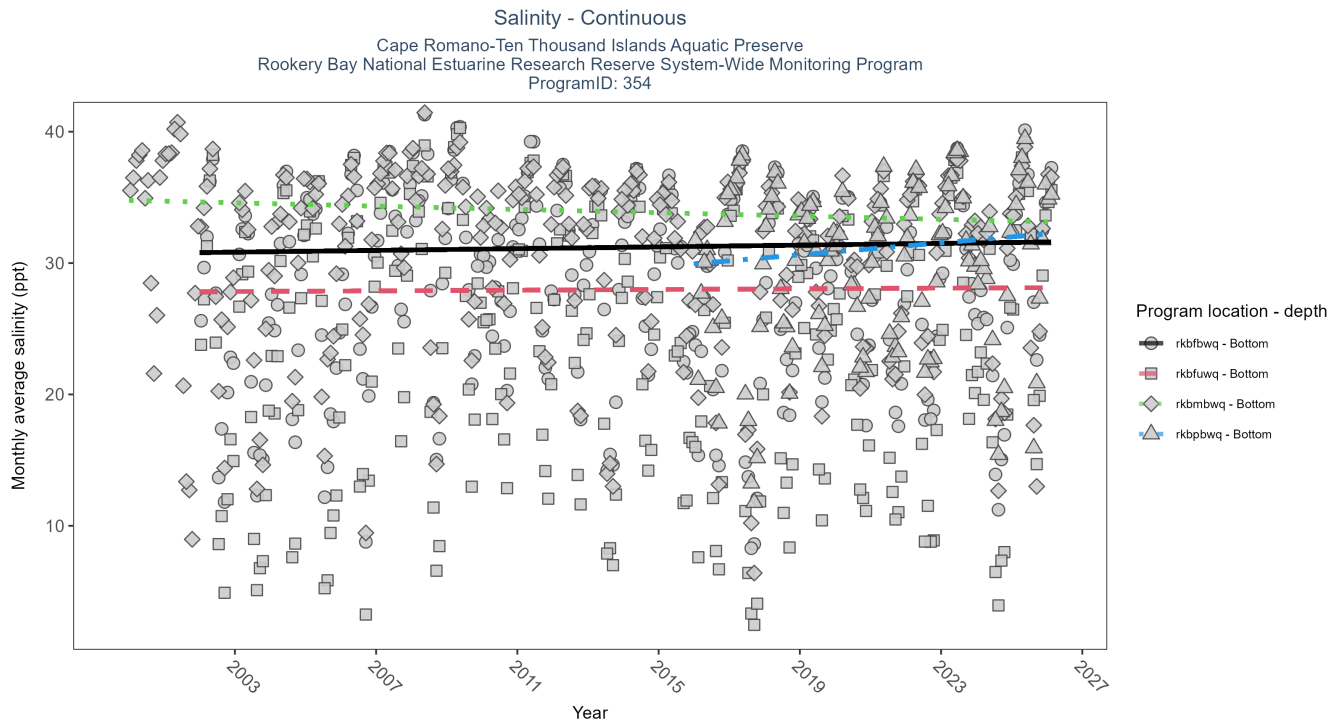


Figure 28: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 32: Seasonal Kendall-Tau Results for Salinity - Program 354

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
rkbfwbq	No significant trend	703182	25	2002 - 2026	29.9	0.04	30.80	0.03	0.34
rkbfuwq	No significant trend	717069	25	2002 - 2026	26.3	0.01	27.82	0.01	0.79
rkmbwbq	Significantly decreasing trend	736129	27	2000 - 2026	33.3	-0.10	34.78	-0.06	0.02
rkbpwbq	No significant trend	333127	11	2016 - 2026	30.5	0.14	29.94	0.23	0.06

At one program location, monthly average salinity increased by 0.31 ppt per year. At two program locations, monthly average salinity decreased by 0.06 ppt per year at one site and by 0.44 ppt per year at the other. No detectable change in monthly average salinity was observed at seven locations. There was insufficient data to fit a model for two locations.

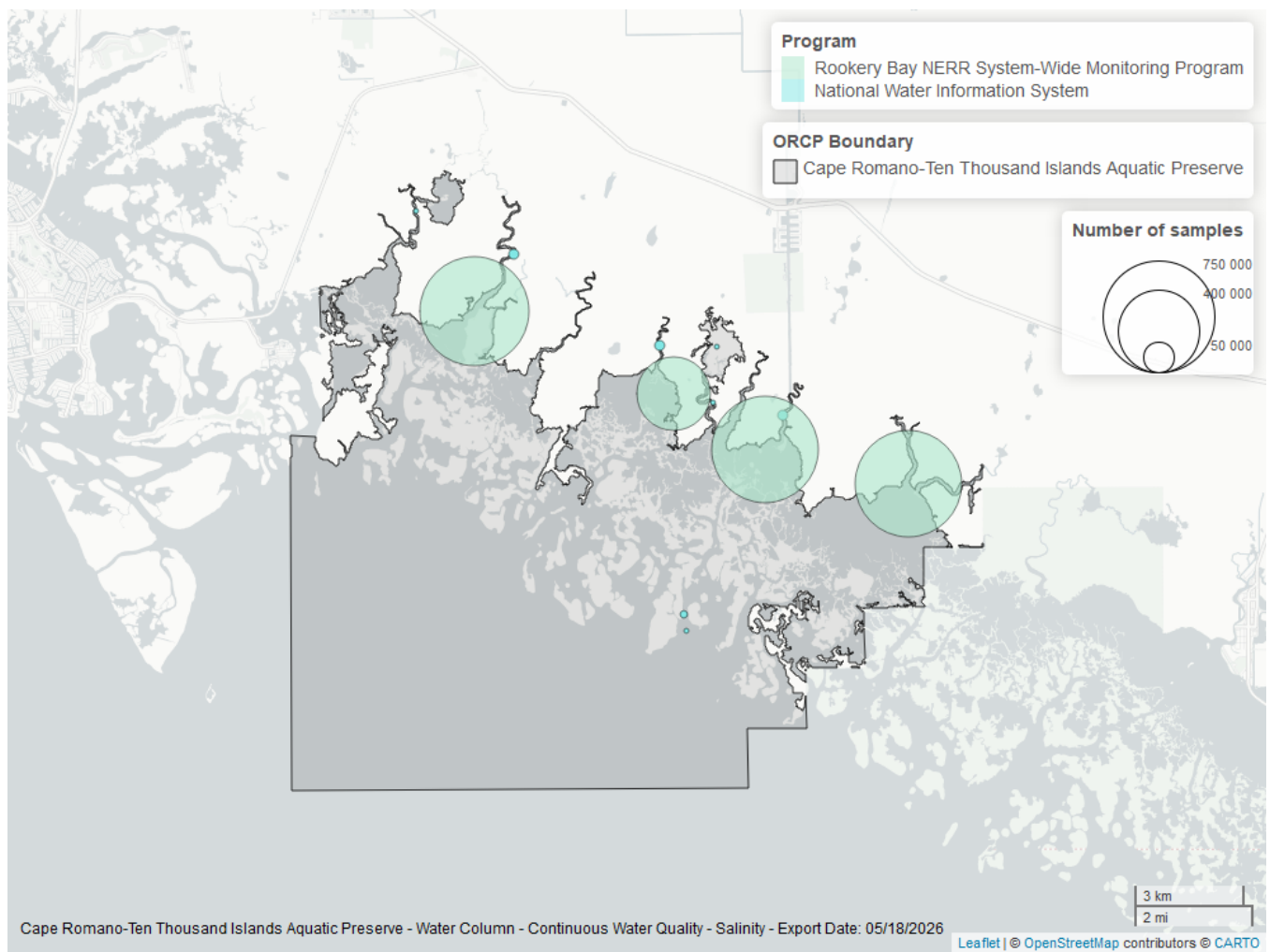


Figure 29: Map showing location of salinity continuous water quality sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Specific Conductivity - Continuous

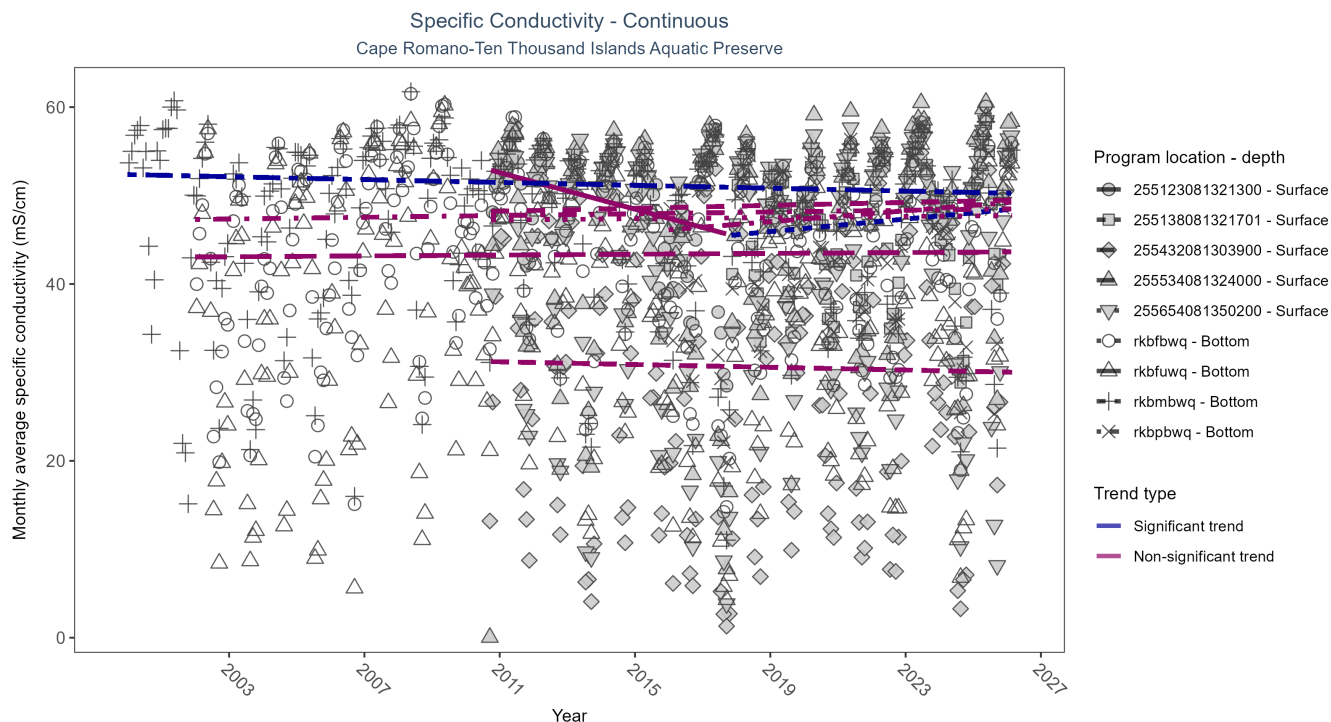


Figure 30: Scatter plot of monthly average specific conductivity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 33: Seasonal Kendall-Tau Results for Specific Conductivity - All Stations

Program Location	Statistical Trend	No. of Samples	No. Years with Data	Period of Record	Median Result Value	Tau	Sen Intercept	Sen Slope	P
255123081321300	No significant trend	1295	6	2010 - 2017	49.10	-0.20	53.63	-1.03	0.20
255138081321701	Significantly increasing trend	2986	10	2017 - 2026	48.30	0.18	45.21	0.36	0.02
255432081303900	No significant trend	5512	17	2010 - 2026	34.40	-0.03	31.28	-0.08	0.61
255534081324000	No significant trend	5509	17	2010 - 2026	48.60	0.04	48.14	0.08	0.43
255654081350200	No significant trend	5547	17	2010 - 2026	48.60	0.03	47.21	0.03	0.56
rkbfbwq	No significant trend	703174	25	2002 - 2026	46.07	0.04	47.30	0.05	0.33
rkbfuwq	No significant trend	717061	25	2002 - 2026	41.17	0.01	43.06	0.02	0.77
rkmbbwq	Significantly decreasing trend	736129	27	2000 - 2026	50.83	-0.09	52.37	-0.08	0.02
rkbpbwq	No significant trend	333114	11	2016 - 2026	47.03	0.13	46.12	0.31	0.07

At one program location, monthly average specific conductivity increased by 0.36 mS/cm per year. At one program location, monthly average specific conductivity decreased by 0.08 mS/cm per year. No detectable change in monthly average specific conductivity was observed at seven locations.

Nekton

The data file used is: **All_NEKTON_Parameters-2026-Jun-04.txt**

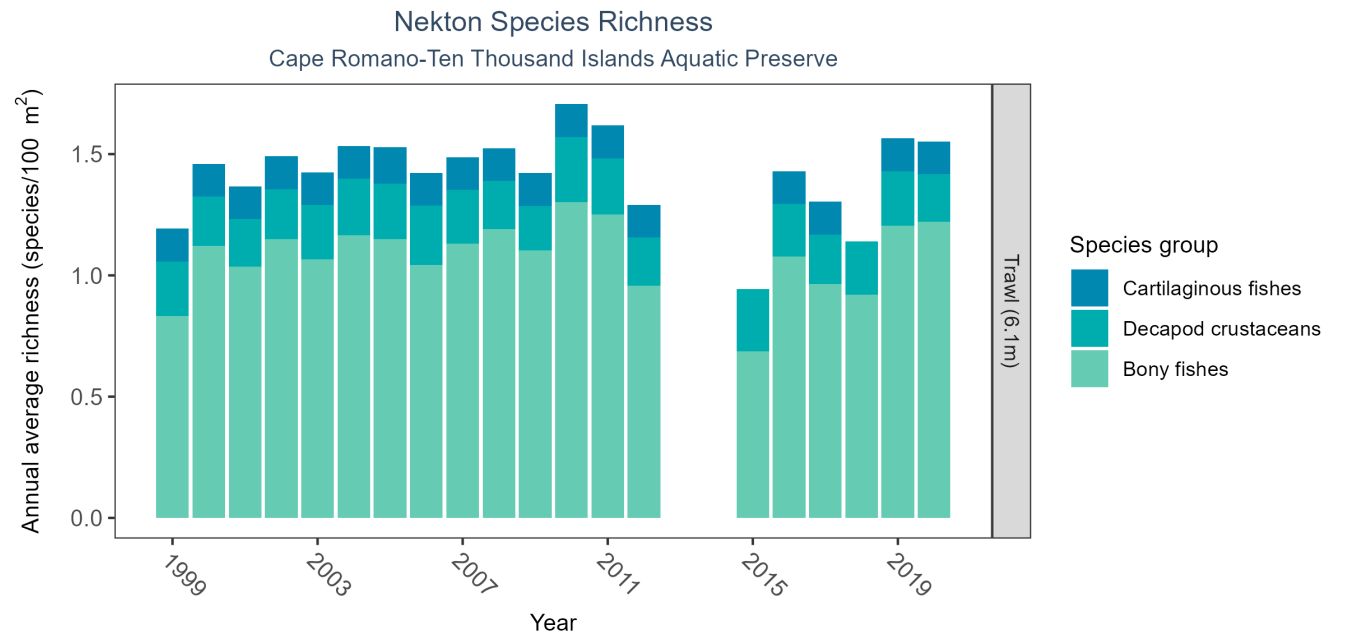


Figure 31: Bar graph(s) of annual average nekton richness over time for species groups occurring in at least 1% of samples. The bar colors represent species groups including bony fishes, cartilaginous fishes, decapod crustaceans (e.g., shrimps, crabs, and lobsters), and cephalopods (e.g., squid). Gear types and sizes are indicated in the panel label.

Table 34: Nekton Species Richness

<i>Gear Type</i>	<i>No. of Samples</i>	<i>No. of Years with Data</i>	<i>Period of Record</i>	<i>Median No. of Taxa</i>	<i>Mean No. of Taxa</i>
Trawl (6.1 m)	5002	20	1999 - 2020	0.4	0.67

The median annual number of taxa was 0.40 based on 5,002 observations collected by 6.1-meter trawl between 1999 and 2020.

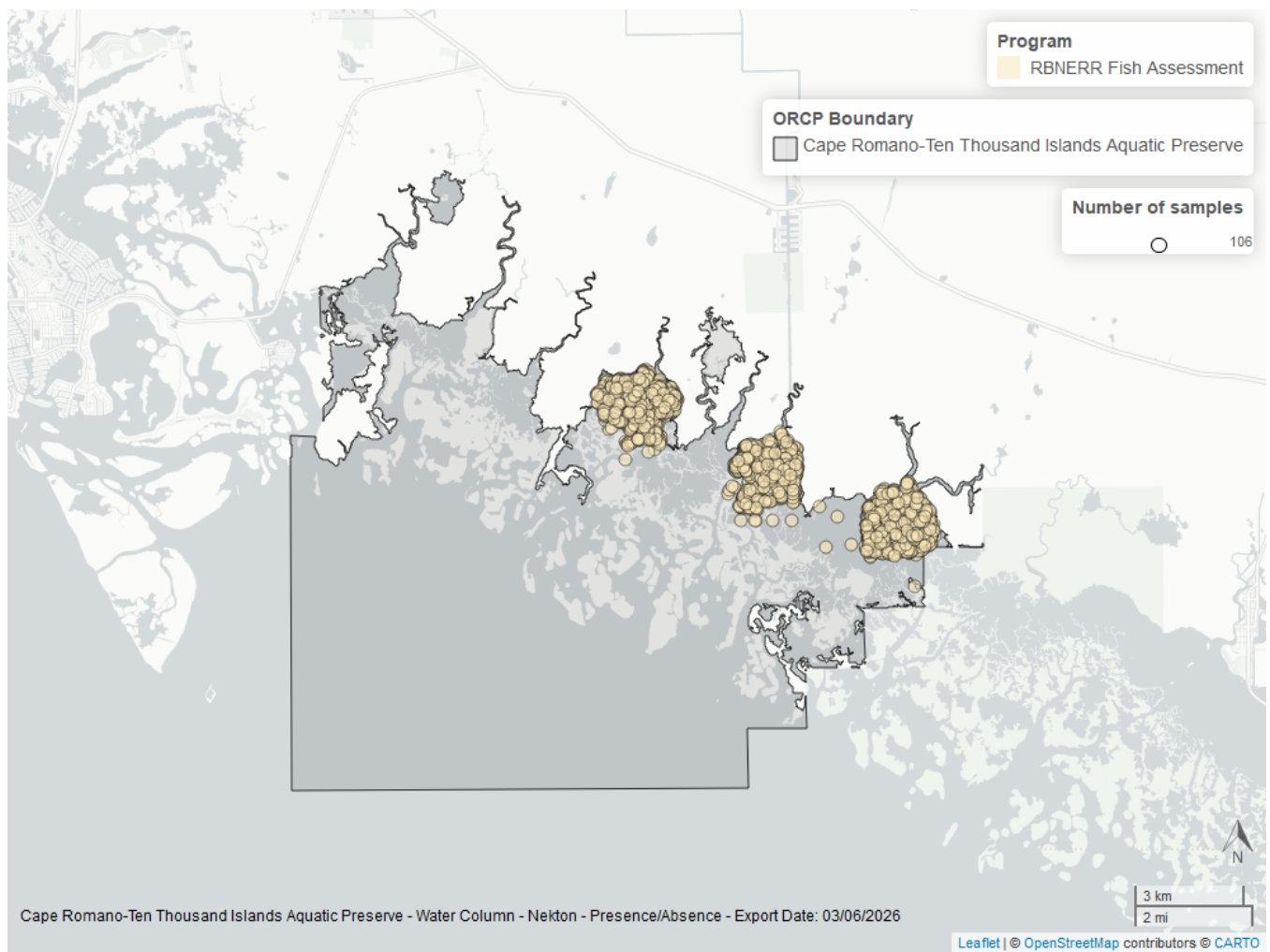


Figure 32: Map showing location of nekton sampling locations within the boundaries of *Cape Romano-Ten Thousand Islands Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

Species list

<i>Acanthostracion quadricornis</i> ¹	<i>Etropus crossotus</i> ¹	<i>Mugil cephalus</i> ¹
<i>Achirus lineatus</i> ¹	<i>Eucinostomus</i> spp. ¹	<i>Mugil</i> spp. ¹
<i>Albula vulpes</i> ¹	<i>Eugerres plumieri</i> ¹	<i>Mycteroperca microlepis</i> ¹
<i>Aluterus schoepfii</i> ¹	<i>Floridichthys carpio</i> ¹	<i>Myrophis punctatus</i> ¹
<i>Anarchopterus criniger</i> ¹	<i>Ginglymostoma cirratum</i> ¹	<i>Nicholsina usta</i> ¹
<i>Anchoa</i> spp. ¹	<i>Gobiesox strumosus</i> ¹	<i>Ogcocephalus cubifrons</i> ¹
<i>Ancylosetta quadrocellata</i> ¹	<i>Gobionellus oceanicus</i> ¹	<i>Ogilbia cayorum</i> ¹
<i>Archosargus probatocephalus</i> ¹	<i>Gobiosoma bosc</i> ¹	<i>Oligoplites saurus</i> ¹
<i>Archosargus rhomboidalis</i> ¹	<i>Gobiosoma robustum</i> ¹	<i>Opisthonema oglinum</i> ¹
<i>Ariopsis felis</i> ¹	<i>Gunterichthys longipenis</i> ¹	<i>Opsanus beta</i> ¹
<i>Bagre marinus</i> ¹	<i>Gymnura micrura</i> ¹	<i>Orthopristis chrysoptera</i> ¹
<i>Bairdiella chrysoura</i> ¹	<i>Harengula jaguana</i> ¹	<i>Paralichthys albigutta</i> ¹
<i>Brevoortia smithi</i> ¹	<i>Hippocampus erectus</i> ¹	<i>Penaeus</i> spp. ¹
<i>Brevoortia</i> spp. ¹	<i>Hippocampus zosterae</i> ¹	<i>Pogonias cromis</i> ¹
<i>Calamus arctifrons</i> ¹	<i>Hoplosternum littorale</i> ¹	<i>Prionotus scitulus</i> ¹
<i>Callinectes sapidus</i> ¹	<i>Hypanus americanus</i> ¹	<i>Prionotus tribulus</i> ¹
<i>Caranx</i> spp. ¹	<i>Hypanus sabinus</i> ¹	<i>Rachycentron canadum</i> ¹
<i>Centropomus undecimalis</i> ¹	<i>Hypostomus plecostomus</i> ¹	<i>Rhinoptera bonasus</i> ¹
<i>Chaetodipterus faber</i> ¹	<i>Hypsoblennius hentz</i> ¹	<i>Sciaenops ocellatus</i> ¹
<i>Chasmodes saburrae</i> ¹	<i>Lagodon rhomboides</i> ¹	<i>Scorpaena brasiliensis</i> ¹
<i>Chilomycterus schoepfii</i> ¹	<i>Leiostomus xanthurus</i> ¹	<i>Selene vomer</i> ¹
<i>Chloroscombrus chrysurus</i> ¹	<i>Lepomis</i> spp. ¹	<i>Serraniculus pumilio</i> ¹
<i>Cichlidae</i> spp. ¹	<i>Leptocephalus larvae</i> ¹	<i>Sphoeroides nephelus</i> ¹
<i>Citharichthys macrops</i> ¹	<i>Lophogobius cyprinoides</i> ¹	<i>Sphoeroides spengleri</i> ¹
<i>Citharichthys spilopterus</i> ¹	<i>Lucania parva</i> ¹	<i>Stephanolepis hispidus</i> ¹
<i>Clupeidae</i> ¹	<i>Lutjanus analis</i> ¹	<i>Symphurus plagiusa</i> ¹
<i>Ctenogobius boleosoma</i> ¹	<i>Lutjanus griseus</i> ¹	<i>Syngnathus louisianae</i> ¹
<i>Ctenogobius smaragdus</i> ¹	<i>Lutjanus synagris</i> ¹	<i>Syngnathus scovelli</i> ¹
<i>Cynoscion arenarius</i> ¹	<i>Menippe mercenaria</i> ¹	<i>Synodus foetens</i> ¹
<i>Cynoscion nebulosus</i> ¹	<i>Menticirrhus</i> spp. ¹	<i>Trachinotus falcatus</i> ¹
<i>Diplectrum formosum</i> ¹	<i>Microgobius gulosus</i> ¹	<i>Trinectes maculatus</i> ¹
<i>Echeneis neucratoides</i> ¹	<i>Microgobius thalassinus</i> ¹	<i>Urophycis floridana</i> ¹
<i>Elops saurus</i> ¹	<i>Micropterus salmoides</i> ¹	<i>Acanthostracion quadricornis</i> ¹
<i>Epinephelus itajara</i> ¹	<i>Monacanthus ciliatus</i> ¹	<i>Achirus lineatus</i> ¹

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